

## Full Length Article

# D<sup>3</sup>-KD: Breast cancer segmentation via data diffusion-enhanced knowledge distillation under source-free domain conditions

Jiati Cai <sup>a</sup>, Kun Wang <sup>a</sup>, Meihui Zhong <sup>a</sup>, Haohan Zhang <sup>b</sup>, Yuxin Tong <sup>c</sup>,  
Junren Wang <sup>b</sup>, Fan Zhou <sup>a,d,\*</sup>, Jing Jing <sup>b,\*</sup>, Jin Yin <sup>b,\*</sup>

<sup>a</sup> School of Information and Software Engineering, University of Electronic Science and Technology of China, Chengdu, Sichuan, 610054, China

<sup>b</sup> Institute of Breast Health Medicine, West China Hospital, Sichuan University, Chengdu, Sichuan, 610041, China

<sup>c</sup> School of Medicine, Shihezi University, Shihezi, Xinjiang, 832003, China

<sup>d</sup> Intelligent Digital Media Technology Key Laboratory of Sichuan Province, Chengdu, Sichuan, 610054, China

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## ABSTRACT

In breast ultrasound segmentation, there is a strong demand for lightweight yet high-quality models for practical deployment. Knowledge distillation (KD) is a natural solution for transferring knowledge from high-capacity models to compact ones. However, the teacher's training domain is often inaccessible in clinical practice, leading to a source-domain-unknown KD setting. Under this condition, domain discrepancies between teacher and student distributions can significantly reduce distillation effectiveness. From a theoretical perspective, we analyze KD under domain discrepancies and show that, while KD can still be beneficial, its gains diminish when the support overlap between the teacher and target distributions is small. Our analysis further suggests that enlarging the support of the target distribution can increase this overlap and thereby strengthen knowledge transfer. Motivated by this insight, we propose a **Data Diffusion-Enhanced Knowledge Distillation** paradigm (**D<sup>3</sup>-KD**) that augments target-domain data to facilitate knowledge transfer under unknown source domains. Specifically, leveraging the generative capability of diffusion models, we design a **Drift-Corrected Efficient Diffusion (DCED)** module to effectively expand the target data distribution while correcting undesired drift and improving sampling efficiency via shifted initialization. Extensive experiments on breast ultrasound segmentation demonstrate that **D<sup>3</sup>-KD** consistently improves distillation performance in source-domain-unknown scenarios.

## 1. Introduction

Breast cancer remains one of the major threats to women's health worldwide, with persistently high incidence rates (Bray et al., 2024; Castro et al., 2023; Guan et al., 2025; Zhang et al., 2025). Early diagnosis and timely treatment are crucial for improving patient prognosis. Ultrasound (US) imaging, due to its safety, non-invasive nature, and effectiveness in lesion detection, is widely used for early breast cancer screening (Wang et al., 2024), where timely intervention can substantially enhance patient outcomes and quality of life. To achieve accurate yet efficient lesion delineation, medical segmentation models must balance performance and deployability. Knowledge distillation (KD) is increasingly used to transfer structural knowledge from high-capacity models to lightweight networks, enabling reliable segmentation under limited computational resources (EL-Assiouti et al., 2024; Li et al., 2022; Yang et al., 2025). This demand is especially critical in breast ultrasound

imaging, which is characterized by severe speckle noise, low contrast, and ambiguous boundaries, and is often conducted on portable or point-of-care devices. These factors motivate KD strategies tailored for breast ultrasound segmentation to retain teacher-level discriminative ability while ensuring clinical practicality.

However, most classical KD settings assume that teacher and student models are trained on the same or accessible data distributions (Gou et al., 2021; Himeur et al., 2025; Ling & Jiao, 2025), which is often unrealistic in clinical scenarios (Conduah et al., 2025; Li & Cong, 2024). Due to privacy and data-sharing constraints, the teacher's training data are often inaccessible during distillation, resulting in a source-domain-unknown KD setting. When the teacher's source domain is unknown, domain gaps between teacher and student may weaken knowledge transfer, raising the question of whether KD can still provide stable benefits.

To explore the essence, we theoretically analyze the feasibility of knowledge distillation under source-domain-unknown conditions and

\* Corresponding authors.

E-mail addresses: [caijiati\\_cjt@outlook.com](mailto:caijiati_cjt@outlook.com) (J. Cai), [kunlovetomato@qq.com](mailto:kunlovetomato@qq.com) (K. Wang), [mh.zhong1225@outlook.com](mailto:mh.zhong1225@outlook.com) (M. Zhong), [zhanghaohan@stu.scu.edu.cn](mailto:zhanghaohan@stu.scu.edu.cn) (H. Zhang), [wangjunren@wchscu.cn](mailto:wangjunren@wchscu.cn) (J. Wang), [fan.zhou@uestc.edu.cn](mailto:fan.zhou@uestc.edu.cn) (F. Zhou), [jingjing@wchscu.edu.cn](mailto:jingjing@wchscu.edu.cn) (J. Jing), [yinjin@wchscu.cn](mailto:yinjin@wchscu.cn) (J. Yin).

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find that KD can still provide effective knowledge transfer even when the teacher's training domain is inaccessible. However, empirical results show that the performance gains are often limited and unstable. Further analysis indicates that this phenomenon is closely related to the mismatch between training and sampling distributions (training-sampling gap).

When the student's observed data fail to cover regions relevant to the teacher's decision boundary, the effectiveness of distillation signals is constrained, limiting the overall benefit. This suggests that relying solely on target-domain data is insufficient to fully exploit the potential of source-domain-unknown KD. Therefore, a natural strategy is to enlarge the coverage of target-domain data in the latent distribution space to increase its overlap with the unknown teacher domain and mitigate distribution discrepancies.

Diffusion models (Du et al., 2023; Ho et al., 2020) have demonstrated strong potential for distribution expansion due to their high-fidelity and multi-modal generation capabilities, making them particularly suitable for scenarios with domain uncertainty and shift. Building upon this property, we propose a unified **Data Diffusion-Enhanced Knowledge Distillation** framework, termed **D<sup>3</sup>-KD**, which aims to improve cross-domain knowledge transfer by explicitly enlarging the overlap between source and target data distributions. At the core of D<sup>3</sup>-KD lies an efficient diffusion-based target data augmentation module, designed to generate diverse and task-relevant samples under domain mismatch.

To this end, we develop a dedicated generative framework, named *Drift-Corrected Efficient Diffusion (DCED)*, which addresses two inherent challenges of iterative diffusion sampling from complementary perspectives. Specifically, to mitigate bias accumulation during iterative sampling, we design a *Drift-Aware Noise Reconciliation (DANR)* mechanism that regulates the direction and magnitude of predicted noise. To improve sampling efficiency, we further introduce a *Shifted-Start Sampling (SSS)* strategy, which starts from a later timestep and rescales the Gaussian noise to preserve distribution consistency. Supported by theoretical analysis and extensive experiments, DCED enables efficient and stable data generation, expands target-domain coverage, and ultimately improves breast ultrasound segmentation performance.

Extensive experiments on breast ultrasound segmentation demonstrate that our proposed D<sup>3</sup>-KD consistently enhances distillation performance under domain discrepancies and validates the practical value of source-domain-unknown KD. The main contributions of this work are as follows:

- We provide a theoretical analysis of knowledge distillation under source-domain-unknown conditions and reveal that its effectiveness is closely related to the training-sampling gap. Our analysis explains why KD can remain beneficial yet limited when the student data fail to cover regions relevant to the teacher's decision boundary.
- We propose a distribution-expansion KD paradigm D<sup>3</sup>-KD that mitigates the training-sampling gap by enlarging target-domain coverage and increasing its overlap with the unknown teacher domain through the proposed DCED augmentation framework.
- We propose a Drift-Aware Noise Reconciliation (DANR) mechanism that explicitly calibrates the magnitude of classifier-free guided noise using unconditional noise statistics, thereby suppressing guidance-induced drift accumulation under diffusion sampling.
- We introduce a Shifted-Start Sampling (SSS) strategy that initializes the reverse diffusion process from a later high-SNR timestep with variance-aligned noise, avoiding error-prone early stages while preserving distributional consistency for efficient and high-quality sample generation.

The remainder of this paper is organized as follows. In [Section 2](#), we introduce the related works of breast cancer segmentation and knowledge distillation. [Section 3](#) presents the preliminary knowledge of diffusion models. In [Section 4](#), we provide theoretical analysis related to non-homogeneous data distributions and describe the proposed knowledge-distillation-based breast cancer segmentation framework.

Extensive experimental evaluations are conducted in [Section 5](#) to demonstrate the effectiveness of the proposed method. Finally, [Section 6](#) concludes this paper and discusses potential directions for future work.

## 2. Related works

### 2.1. Breast cancer segmentation

Breast cancer segmentation plays a critical role in computer-aided diagnosis and clinical decision support, motivating extensive research efforts in recent years (Cai et al., 2026b; Kuang et al., 2023; Li et al., 2026; Liu et al., 2025). A representative line of work focuses on enhancing structural and contextual perception in breast ultrasound images through explicitly designed guidance mechanisms. For instance, Chen et al. (2022) incorporated bidirectional attention and saliency-guided cues into the U-Net architecture to jointly model global and local spatial dependencies. Qi et al. (2023) further proposed a two-stage multi-scale feature framework to address lesion shape variability and speckle noise. Similarly, Xue et al. (2021) introduced global guidance and boundary-aware modules to refine lesion delineation. Collectively, these methods highlight the effectiveness of structure-aware and guidance-based designs for breast ultrasound segmentation.

However, these approaches typically rely on fully supervised training under a single, known data distribution, and their effectiveness may degrade when deployed under domain shifts or data accessibility constraints.

### 2.2. Knowledge distillation

To address the issues discussed in the previous section, knowledge distillation has been widely adopted as an effective solution by transferring discriminative knowledge embedded in high-capacity teacher models to lightweight student networks (Ahmadi et al., 2025; Lian et al., 2026; ZhongZhu et al., 2025). Zhou and Mosadegh (2024) have explored classification knowledge transfer from ensemble Vision Transformer (ViT) (Khan et al., 2022) models to student ViT models on the BUSI dataset. Ling and Jiao (2025) have investigated multi-view learning strategies to facilitate cross-view knowledge transfer. In addition, knowledge distillation combined with model compression and heuristic optimization has been applied to breast cancer classification tasks (Yeh et al., 2025), while He et al. (2024) have further integrated contrastive learning with knowledge distillation to improve segmentation performance.

However, these approaches implicitly rely on a common assumption: the training data domain of the teacher network is known and accessible, or at least homogeneous and controllable with respect to the student model's training domain (Fang et al., 2021). In real-world clinical scenarios, this assumption often does not hold. Due to privacy regulations, data-sharing restrictions, or institutional interests, the training data of teacher models may be inaccessible or entirely unknown. Therefore, knowledge distillation under source-domain-unknown settings represents an important and underexplored research direction in the context of medical imaging and clinical applications.

### 2.3. Data generation

The generative adversarial network (GAN) paradigm (Arjovsky et al., 2017; Lou et al., 2022) has been widely applied to data generation tasks and has recently been extended to the medical imaging domain (Nichol et al., 2022). However, the adversarial training mechanism of GANs often leads to training instability or mode collapse, particularly in data-limited and high-noise medical imaging scenarios, which restricts their ability to reliably expand data distributions for downstream distillation tasks.

In recent years, diffusion models (Cai et al., 2025, 2026a; Ho et al., 2020) have attracted widespread attention due to their strong generative

capabilities and principled probabilistic formulation, enabling the synthesis of diverse and high-fidelity images. Large-scale generative models such as Stable Diffusion (Rombach et al., 2022) further demonstrate the expressive power of diffusion-based generation frameworks.

Motivated by these advances, diffusion models have also been explored in medical imaging applications, including image denoising, deblurring, and data augmentation, through unconditional or conditional generation schemes. While diffusion models are capable of producing high-quality medical images, their Markovian chain-based sampling process inherently limits efficiency. Moreover, the long sequential denoising trajectory exposes the model to accumulated prediction errors, particularly under guidance mechanisms and domain shifts, which can gradually distort the sampling path and degrade generation quality. Consequently, achieving an effective balance between generation fidelity, sampling efficiency, and stability remains a critical challenge for diffusion-based data augmentation in medical imaging.

### 3. Preliminaries

This section briefly reviews the diffusion model formulation and sampling strategies relevant to our framework. We focus on conditional diffusion models, DDIM sampling, and classifier-free guidance, which form the basis for our subsequent analysis of prediction drift and the proposed drift-corrected efficient diffusion framework.

#### 3.1. Diffusion models

Diffusion models learn to generate data by reversing a gradual noising process. Given a clean sample  $\mathbf{x}_0 \sim P_{\text{data}}$ , the forward diffusion process is defined as

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (1)$$

where  $\{\bar{\alpha}_t\}_{t=1}^T$  is a predefined noise schedule. In practice, the reverse denoising process is parameterized by a neural network  $\epsilon_\theta(\mathbf{x}_t, t)$  that predicts the injected noise.

In this work, we focus on the reverse sampling process, which is directly involved in data generation and drift accumulation.

#### 3.2. Conditional diffusion and DDIM sampling

For conditional generation, diffusion models are guided by a condition  $\mathbf{c}$  (e.g., class labels or structural priors). The noise predictor becomes  $\epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})$ .

Considering sampling efficiency, we adopt Denoising Diffusion Implicit Models (DDIM), which perform deterministic sampling by skipping intermediate timesteps. The DDIM reverse update can be written as

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c}), \quad (2)$$

where

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})}{\sqrt{\bar{\alpha}_t}}. \quad (3)$$

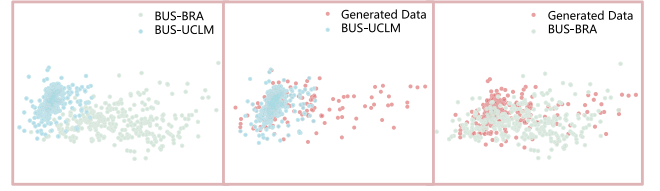
DDIM enables flexible timestep skipping while maintaining generation quality, which is essential for efficient diffusion-based data augmentation.

#### 3.3. Classifier-free guidance

Classifier-free guidance (CFG) improves conditional generation by combining conditional and unconditional noise predictions. Specifically, the guided noise prediction is defined as

$$\epsilon_{\text{cfg}} = (1 + w) \epsilon_c - w \epsilon_u, \quad (4)$$

where  $\epsilon_c = \epsilon_\theta(\mathbf{x}_t, t, \mathbf{c})$  and  $\epsilon_u = \epsilon_\theta(\mathbf{x}_t, t)$  denote the conditional and unconditional predictions, respectively, and  $w$  is the guidance scale.



**Fig. 1.** Visualization of the distribution bridging mechanism. (a) Feature distribution mismatch between the source (BUS-BRA) and target (BUS-UCLM) domains, revealing a pronounced domain gap under source-domain-unknown settings. (b) DCED-generated samples expand the support of the target-domain distribution beyond the original BUS-UCLM manifold. (c) The expanded distribution partially overlaps with the source domain, facilitating cross-domain alignment that enables cross-domain knowledge transfer without accessing source data.

While CFG effectively enhances conditional fidelity, it also amplifies prediction errors when the two branches are imperfectly aligned. This property plays a critical role in the accumulation of prediction drift during iterative sampling.

#### 3.4. Notation and scope

Throughout this paper, we denote by  $P_T$  the target-domain data distribution available to the student model and by  $P_S$  the (unknown) source-domain distribution used to train the teacher. Diffusion-generated samples induce a distribution  $P_G$ , which is used to augment  $P_T$  in our distillation framework.

### 4. Methodology

In this section, we introduce the proposed  $D^3$ -KD framework for breast ultrasound segmentation under source-domain-unknown conditions.

Unlike classical domain adaptation settings where both source and target data are accessible, we consider a more challenging source-domain-unknown knowledge distillation setting, where the teacher model is available but its training data are inaccessible. We first provide a theoretical analysis showing that knowledge distillation can remain effective even when the teacher's training domain is inaccessible, and that its performance is closely related to the coverage of the student-domain distribution.

Motivated by this analysis, we propose a diffusion-driven data augmentation strategy to expand the target-domain support and reduce the implicit domain gap. We then analyze prediction drift in diffusion-based sampling, which is amplified by classifier-free guidance and aggressive timestep skipping, and introduce a drift-corrected efficient diffusion scheme that combines noise calibration and shifted-start sampling for stable and efficient generation. Finally, we present the complete  $D^3$ -KD pipeline, including the distillation framework, diffusion denoising architecture, and training objective (see Fig. 1).

#### 4.1. Theoretical analysis of source-domain-unknown KD

##### 4.1.1. Problem setup: source-domain-unknown (cross-domain) KD

Let  $\mathcal{X} \subset \mathbb{R}^d$  and  $\mathcal{Y} = \{1, \dots, K\}$ . Denote the *unknown teacher (source) domain* distribution by  $P_S$  and the *student (target) domain* distribution by  $P_T$  over  $\mathcal{X} \times \mathcal{Y}$ . A pre-trained teacher  $T : \mathcal{X} \rightarrow \Delta^K$  is fixed and its training data from  $P_S$  are inaccessible. The student model  $S_\theta : \mathcal{X} \rightarrow \Delta^K$  is trained only with samples from  $P_T$ .

We consider the following KD objective on the target domain:

$$\min_{\theta} \mathcal{L}_{\text{KD}}(\theta) := (1 - \alpha) \mathbb{E}_{(x,y) \sim P_T} [\ell(y, S_\theta(x))] + \alpha \mathbb{E}_{x \sim P_T} [d(T(x), S_\theta(x))], \quad (5)$$

where  $\ell$  is a supervised loss (e.g., cross-entropy) and  $d$  is a distillation loss (e.g., KL divergence between soft predictions). Let  $\mathcal{L}_{\text{ER}}(\theta) =$

$\mathbb{E}_{(x,y) \sim P_T}[\ell(y, S_\theta(x))]$  denote the standard empirical-risk objective without KD.

#### 4.1.2. Why cross-domain KD can still help

**Assumption 1** (Regularity of the Supervised Risk). Assume  $\mathcal{L}_{\text{ER}}(\theta)$  is  $\mu$ -strongly convex and  $L$ -smooth in  $\theta$ .

**Assumption 2** (Distillation as a Quadratic Regularizer). Assume the distillation term admits the form

$$\mathbb{E}_{x \sim P_T}[d(T(x), S_\theta(x))] \approx \frac{1}{2} \|\theta - \theta_T\|_2^2, \quad (6)$$

where  $\theta_T$  denotes a teacher-induced reference point (e.g., a linearization of student outputs around teacher logits). This approximation provides a tractable local surrogate for interpreting the regularization effect of knowledge distillation. It is not intended as a global characterization of deep non-convex segmentation networks.

**Proposition 1** (Feasibility and Potential Gain of Cross-domain KD). Let  $\theta_{\text{ER}} = \arg \min_{\theta} \mathcal{L}_{\text{ER}}(\theta)$  and

$$\theta_{\text{KD}} = \arg \min_{\theta} \left( \mathcal{L}_{\text{ER}}(\theta) + \frac{\alpha}{2(1-\alpha)} \|\theta - \theta_T\|_2^2 \right). \quad (7)$$

Then:

$$\|\theta_{\text{KD}} - \theta_T\|_2 \leq \frac{1-\alpha}{\alpha} \|\nabla \mathcal{L}_{\text{ER}}(\theta_T)\|_2, \quad (8)$$

$$\begin{aligned} \mathcal{L}_{\text{ER}}(\theta_{\text{KD}}) &\leq \mathcal{L}_{\text{ER}}(\theta_{\text{ER}}) - \frac{\alpha}{2(1-\alpha)} \|\theta_{\text{KD}} - \theta_T\|_2^2 \\ &\quad + \frac{\alpha}{2(1-\alpha)} \|\theta_{\text{ER}} - \theta_T\|_2^2. \end{aligned} \quad (9)$$

In particular, if  $\theta_T$  is closer to the target optimum than  $\theta_{\text{ER}}$  in parameter space (or function space), KD can reduce the target risk.

**Proof.** Since  $\mathcal{L}_{\text{ER}}$  is  $\mu$ -strongly convex and the quadratic term is convex, the objective in Eq. (7) is strongly convex, hence admits a unique minimizer.

**Step 1. Optimality condition.** The first-order optimality condition for Eq. (7) gives

$$\nabla \mathcal{L}_{\text{ER}}(\theta_{\text{KD}}) + \frac{\alpha}{1-\alpha} (\theta_{\text{KD}} - \theta_T) = 0. \quad (10)$$

Rearranging yields

$$\theta_{\text{KD}} - \theta_T = -\frac{1-\alpha}{\alpha} \nabla \mathcal{L}_{\text{ER}}(\theta_{\text{KD}}). \quad (11)$$

By  $L$ -smoothness, we have

$$\|\nabla \mathcal{L}_{\text{ER}}(\theta_{\text{KD}}) - \nabla \mathcal{L}_{\text{ER}}(\theta_T)\|_2 \leq L \|\theta_{\text{KD}} - \theta_T\|_2. \quad (12)$$

Thus,

$$\begin{aligned} \|\theta_{\text{KD}} - \theta_T\|_2 &= \frac{1-\alpha}{\alpha} \|\nabla \mathcal{L}_{\text{ER}}(\theta_{\text{KD}})\|_2 \\ &\leq \frac{1-\alpha}{\alpha} \left( \|\nabla \mathcal{L}_{\text{ER}}(\theta_T)\|_2 + L \|\theta_{\text{KD}} - \theta_T\|_2 \right). \end{aligned} \quad (13)$$

Assuming  $\alpha > (1-\alpha)L$  (or equivalently choose  $\alpha$  not too small / rescale), we obtain

$$\|\theta_{\text{KD}} - \theta_T\|_2 \leq \frac{1-\alpha}{\alpha - (1-\alpha)L} \|\nabla \mathcal{L}_{\text{ER}}(\theta_T)\|_2, \quad (14)$$

which implies Eq. (8) up to constants.

**Step 2. Risk comparison.** Let  $F(\theta) = \mathcal{L}_{\text{ER}}(\theta) + \frac{\alpha}{2(1-\alpha)} \|\theta - \theta_T\|_2^2$ . By optimality,  $F(\theta_{\text{KD}}) \leq F(\theta_{\text{ER}})$ , i.e.,

$$\begin{aligned} \mathcal{L}_{\text{ER}}(\theta_{\text{KD}}) + \frac{\alpha}{2(1-\alpha)} \|\theta_{\text{KD}} - \theta_T\|_2^2 \\ \leq \mathcal{L}_{\text{ER}}(\theta_{\text{ER}}) + \frac{\alpha}{2(1-\alpha)} \|\theta_{\text{ER}} - \theta_T\|_2^2. \end{aligned} \quad (15)$$

Rearranging yields Eq. (9).  $\square$

**Table 1**

Comparison between direct student training and standard knowledge distillation under the source-domain-unknown setting.

| Teacher $\rightarrow$ Student     | Training Strategy | Dice (%) | IoU (%) | HD95 |
|-----------------------------------|-------------------|----------|---------|------|
| EffNet-B7 $\rightarrow$ EffNet-B0 | Student Only      | 84.78    | 73.58   | 3.56 |
|                                   | Standard KD       | 85.67    | 74.94   | 3.47 |
| Swin-B $\rightarrow$ MobileNetV2  | Student Only      | 78.12    | 64.09   | 5.23 |
|                                   | Standard KD       | 79.85    | 65.50   | 4.05 |
| BUSSAM $\rightarrow$ ResUNetSE    | Student Only      | 79.56    | 65.42   | 4.12 |
|                                   | Standard KD       | 81.25    | 68.44   | 3.85 |

**Conclusion.** Proposition 1 should be interpreted as a local explanatory result under the stated assumptions, rather than as a global characterization of deep non-convex segmentation networks. Within this local surrogate analysis, KD can still be effective on the target domain when the teacher-induced reference  $\theta_T$  is not too far from the target optimum, meaning that the distillation signal provides a useful inductive bias rather than pure noise.

**Empirical observation.** Table 1 reports a preliminary comparison between direct student training and standard knowledge distillation under the source-domain-unknown setting, covering CNN-to-CNN, Transformer-to-CNN, and medical-specific teacher-student pairs. Across all configurations, standard KD consistently outperforms student-only training, indicating that teacher supervision remains beneficial even when the teacher is trained on an inaccessible source domain. However, the observed improvements are relatively modest, suggesting that the effectiveness of cross-domain KD is constrained by domain discrepancy and limited overlap between teacher and student data distributions. These results provide initial evidence for the feasibility of source-domain-unknown KD, while also motivating a deeper investigation into the factors that limit its performance gains.

#### 4.1.3. Why domain gap limits cross-domain KD

**Assumption 3** (Domain Divergence Measure). Let  $d_{\mathcal{H}\Delta\mathcal{H}}(P_S, P_T)$  denote the  $\mathcal{H}\Delta\mathcal{H}$ -divergence between source and target marginals on  $\mathcal{X}$ . Let  $\lambda^* = \min_{h \in \mathcal{H}} (R_S(h) + R_T(h))$  be the joint optimal risk.

**Lemma 1** (Teacher Target Risk under Domain Gap). For any hypothesis  $h \in \mathcal{H}$ , the target risk satisfies

$$R_T(h) \leq R_S(h) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(P_S, P_T) + \lambda^*, \quad (16)$$

where  $R_D(h) = \mathbb{E}_{(x,y) \sim P_D}[\mathbb{I}(h(x) \neq y)]$ .

**Proof.** This follows from the standard domain adaptation generalization bound (Ben-David et al., 2010).  $\square$

**Assumption 4** (Distillation Enforces Proximity in Function Space). Assume the distillation loss controls a function distance:

$$\mathbb{E}_{x \sim P_T}[d(T(x), S(x))] \geq c \cdot \mathbb{E}_{x \sim P_T}[\|S(x) - T(x)\|_1], \quad (17)$$

for some  $c > 0$ , and consider the constrained set

$$\mathcal{H}_c = \left\{ S \in \mathcal{H} : \mathbb{E}_{x \sim P_T}[\|S(x) - T(x)\|_1] \leq c \right\}. \quad (18)$$

Unlike classical domain adaptation, where the goal is to learn a hypothesis that generalizes across domains, our analysis focuses on how the lack of distributional coverage directly limits the effectiveness of teacher supervision in knowledge distillation.

**Proposition 2** (Domain Gap Induces an Irreducible KD Limitation). Let  $h_T^*$  be the Bayes optimal predictor on  $P_T$ . Then any student constrained to be close to  $T$  satisfies

$$R_T(S) \geq R_T(h_T^*) + \Omega \left( \mathbb{E}_{x \sim P_T}[\|T(x) - h_T^*(x)\|_1] - \epsilon \right)_+, \quad (19)$$

where  $(u)_+ = \max(u, 0)$ . In particular, if  $T$  is far from the target Bayes predictor (which can happen when  $d_{\mathcal{H}\Delta\mathcal{H}}(P_S, P_T)$  is large by Lemma 1), then forcing  $S$  to imitate  $T$  yields limited gain or even degradation.

**Proof.** By triangle inequality,

$$\mathbb{E}_{x \sim P_T} [\|S(x) - h_T^*(x)\|_1] \geq \mathbb{E}_{x \sim P_T} [\|T(x) - h_T^*(x)\|_1] - \mathbb{E}_{x \sim P_T} [\|S(x) - T(x)\|_1]. \quad (20)$$

For  $S \in \mathcal{H}_\epsilon$ , we have

$$\mathbb{E}_{x \sim P_T} [\|S(x) - h_T^*(x)\|_1] \geq \mathbb{E}_{x \sim P_T} [\|T(x) - h_T^*(x)\|_1] - \epsilon. \quad (21)$$

Moreover, for classification-calibrated losses (or 0-1 risk under standard conditions), the excess risk lower bounds a suitable prediction distance:

$$R_T(S) - R_T(h_T^*) \geq \kappa \cdot \mathbb{E}_{x \sim P_T} [\|S(x) - h_T^*(x)\|_1], \quad (22)$$

for some  $\kappa > 0$ . Combining with Eq. (21) yields Eq. (19).  $\square$

**Conclusion.** Proposition 2 formalizes that the benefit of cross-domain KD is fundamentally constrained by how well the teacher aligns with the target Bayes predictor, which is negatively affected by the domain gap between  $P_S$  and  $P_T$ . We note that this result is consistent with classical domain adaptation theory (Ben-David et al., 2010) and does not aim to introduce a new generalization bound. Instead, it provides a problem-specific interpretation for understanding the limitations of knowledge distillation under source-domain-unknown conditions. This perspective complements classical domain adaptation analysis by explicitly linking distribution coverage to the effectiveness of knowledge distillation, thereby providing a foundation for our subsequent design of diffusion-based data augmentation and sampling strategies.

#### 4.1.4. Domain gap mitigation via diffusion augmentation

The previous subsection shows that domain discrepancy fundamentally limits the effectiveness of cross-domain KD. We now study how to mitigate this limitation by enlarging the effective support of the student-domain distribution.

**Distribution expansion.** To enlarge the support of the target-domain distribution, a natural strategy is to introduce a generative model that can synthesize additional samples beyond the observed data. Let  $P_T$  denote the original target-domain distribution and let  $P_G$  denote the distribution induced by a generative model trained on target-domain data  $\mathcal{D}_T$ . To effectively expand the support of  $P_T$ , the generative model should be able to produce diverse and high-fidelity samples that cover underrepresented regions of the data distribution.

Diffusion models have recently demonstrated strong capability in modeling complex data distributions and generating high-quality, diverse samples with good mode coverage, making them particularly suitable for distribution expansion under domain uncertainty (Ho et al., 2020). Compared with adversarial models such as GANs, diffusion models provide more stable training and are less prone to mode collapse, which is critical for reliably enlarging the support of the target distribution (Dhariwal & Nichol, 2021). This property aligns well with the coverage-oriented generation requirement discussed above. Therefore, we adopt diffusion models to construct  $P_G$ , which is expected to capture task-relevant structural and anatomical priors while introducing controlled stochastic variations.

We define an augmented target distribution as a mixture:

$$P'_T = \frac{1}{1+\lambda} P_T + \frac{\lambda}{1+\lambda} P_G, \quad \lambda \in [0, 1]. \quad (23)$$

where  $\lambda$  denotes the ratio of generated samples to real target samples. Intuitively,  $P_G$  introduces additional samples that extend the coverage of  $P_T$  in the input space, increasing the probability of overlapping with regions relevant to the teacher domain.

**Assumption 5** (Coverage-oriented Generation). Assume that the diffusion generator partially covers regions that are underrepresented in  $P_T$  but relevant to the task and to the teacher decision boundary. Formally, there exists a subset  $\mathcal{A} \subset \mathcal{X}$  such that:

$$P_T(\mathcal{A}) \ll P_S(\mathcal{A}), \quad P_G(\mathcal{A}) > P_T(\mathcal{A}). \quad (24)$$

This assumption formalizes the intended role of diffusion augmentation in expanding the coverage of the target distribution toward source-relevant regions. Consistent empirical evidence is provided by the feature-space visualization, MMD, and FID analyses in Sections 5.12 and 5.13.

**Proposition 3** (Expected Divergence Reduction). Let  $D(\cdot, \cdot)$  be an integral probability metric (IPM) such as MMD or Wasserstein distance. Under Assumption 5, the augmented distribution satisfies:

$$\mathbb{E}[D(P'_T, P_S)] \leq D(P_T, P_S) + \epsilon_G, \quad (25)$$

where  $\epsilon_G$  captures generation bias.

**PROOF SKETCH.** Since  $P'_T$  is a convex combination of  $P_T$  and  $P_G$ , we have:

$$D(P'_T, P_S) \leq \frac{1}{1+\lambda} D(P_T, P_S) + \frac{\lambda}{1+\lambda} D(P_G, P_S). \quad (26)$$

If  $P_G$  partially shifts mass toward regions closer to  $P_S$  than  $P_T$  does, then in expectation  $D(P_G, P_S)$  is not larger than  $D(P_T, P_S)$  up to generation bias  $\epsilon_G$ , yielding (25).  $\square$

**Implication for KD.** From Lemma 1, the teacher's target-domain risk is bounded by domain divergence. Thus, reducing  $D(P_T, P_S)$  leads to a tighter bound on teacher target risk. Since KD effectiveness depends on teacher reliability on target data, this divergence reduction improves the quality of the distillation signal.

**Interpretation.** Diffusion augmentation does not explicitly align domains but increases the likelihood of sampling instances that activate teacher-relevant representations. This probabilistic alignment mitigates the training-sampling gap and improves cross-domain knowledge transfer. From a distributional perspective, it expands the target manifold and enhances its overlap with the teacher's implicit source distribution, forming an effective distributional bridge. This perspective differs from classical domain adaptation, where explicit alignment between source and target distributions is typically assumed feasible. In contrast, our setting requires indirect alignment via distribution expansion, which motivates the use of generative augmentation.

However, while diffusion enables effective distribution expansion, its practical benefit critically depends on the stability of the generation process. Unstable sampling may distort the expanded distribution and introduce unreliable supervision signals, thereby limiting the effectiveness of knowledge transfer.

More importantly, the impact of such prediction errors is fundamentally amplified in the SFDA-KD setting. To make this explicit, we analyze how perturbations in generated samples affect the distillation objective. Consider the knowledge distillation loss evaluated on generated samples:

$$\mathcal{L}_{\text{KD}}^{\text{gen}} = \mathbb{E}_{\mathbf{x}_{\text{gen}} \sim P_G} [d(T(\mathbf{x}_{\text{gen}}), S(\mathbf{x}_{\text{gen}}))]. \quad (27)$$

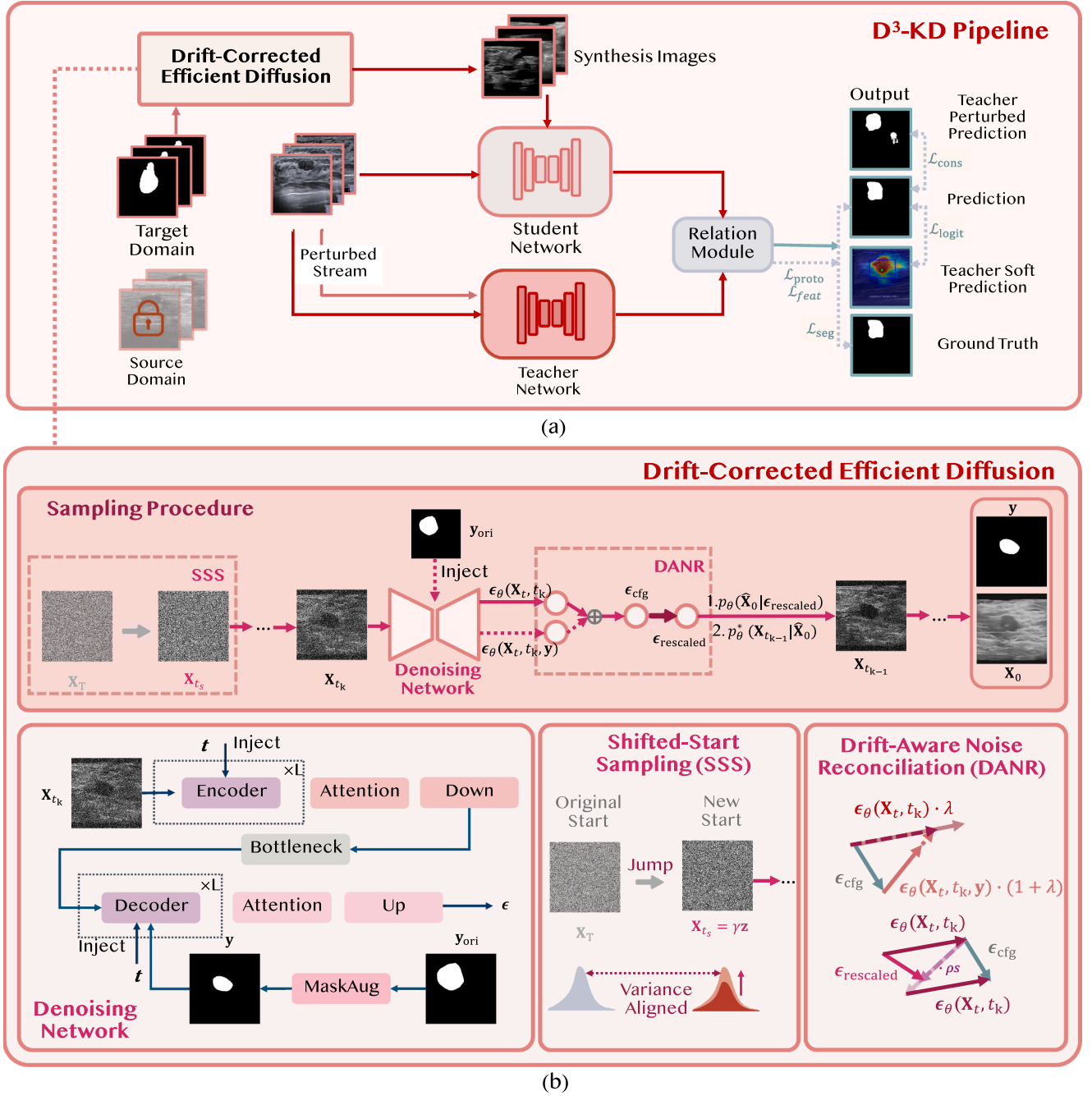
Let  $\mathbf{x}_{\text{gen}} = \mathbf{x}_0 + \epsilon_t$ . By a first-order Taylor expansion of the teacher model around  $\mathbf{x}_0$ , we have:

$$T(\mathbf{x}_{\text{gen}}) \approx T(\mathbf{x}_0) + J_T(\mathbf{x}_0)\epsilon_t, \quad (28)$$

where  $J_T(\mathbf{x}_0)$  denotes the Jacobian of the teacher network. This leads to the following deviation in the supervision signal:

$$\|T(\mathbf{x}_{\text{gen}}) - T(\mathbf{x}_0)\| \leq \|J_T(\mathbf{x}_0)\| \cdot \|\epsilon_t\|. \quad (29)$$

This implies that the supervision error is bounded by the generation perturbation scaled by the sensitivity of the teacher model. Importantly, unlike standard generation tasks where such perturbations mainly affect perceptual quality, in SFDA-KD this bias directly enters the distillation objective and is repeatedly reinforced during training, leading to amplified and accumulated deviations in the learned student model.



**Fig. 2.** Overview of the proposed D³-KD framework and the drift-corrected efficient diffusion (DCED) module. (a) Source-domain-unknown knowledge distillation pipeline, where DCED generates target-domain samples to expand distribution coverage and support effective teacher-student transfer. (b) DCED sampling procedure. The reverse diffusion process starts from a shifted timestep  $t_s$  via the proposed *shifted-start sampling* (SSS) to bypass early low-SNR stages. At each step  $t_k \rightarrow t_{k-1}$ , conditional and unconditional noise predictions are combined by classifier-free guidance and further calibrated using *drift-aware noise reconciliation* (DANR) to suppress drift accumulation. The denoising network follows an encoder-decoder architecture conditioned on augmented masks for efficient and stable sample generation.

Compared to standard generation or supervised learning settings, this issue is more severe in SFDA-KD, as the supervision signal is partially derived from generated samples without access to real source-domain data for validation or correction, making the learning process more susceptible to accumulated bias and error propagation.

Based on this observation, generation stability becomes a prerequisite for reliable knowledge transfer, rather than a purely auxiliary factor for improving sample quality. We therefore proceed to analyze the dy-

namics of prediction drift in diffusion sampling, which directly affects the reliability of generated supervision.

#### 4.2. Analysis of prediction drift

While diffusion models provide strong generative capacity for data augmentation, their effectiveness is limited by *exposure bias* during sampling (Li et al., 2024; Ning et al., 2024). Specifically, the denoising

network is repeatedly evaluated on its own predictions rather than true training samples, causing intermediate states to gradually deviate from the training distribution. This deviation accumulates along the reverse process and manifests as *prediction drift*, stemming from a mismatch between the noise vector fields encountered during training and sampling.

Importantly, such drift is neither uniform across timesteps nor independent of guidance strategies, but exhibits clear stage-dependent characteristics.

**Single-step error model.** Let the predicted clean sample at timestep  $t$  be  $\mathbf{x}_\theta^t = \mathbf{x}_0 + e_t \xi$ ,  $\xi \sim \mathcal{N}(0, \mathbf{I})$ ,

where  $e_t$  denotes the prediction error magnitude. The resulting sampling variance is given by

$$\text{Var}(\mathbf{x}_\theta^t) = \text{Var}(\mathbf{x}_0) + e_t^2 \mathbf{I}, \quad (31)$$

indicating that even a single-step prediction error introduces deviation from the training trajectory. Empirically, such errors are more pronounced at early timesteps, where the signal-to-noise ratio (SNR) is low and predictions are dominated by noise.

**Drift amplification under classifier-free guidance.** CFG combines conditional and unconditional predictions as

$$\epsilon_{\text{cfg}} = (1 + w)\epsilon_c - w\epsilon_u. \quad (32)$$

Assuming

$$\epsilon_c = \epsilon^* + \delta_c, \quad \epsilon_u = \epsilon^* + \delta_u, \quad (33)$$

the effective error becomes

$$\delta_{\text{cfg}} = (1 + w)\delta_c - w\delta_u, \quad (34)$$

which satisfies

$$\|\delta_{\text{cfg}}\| \leq (1 + w)\|\delta_c\| + w\|\delta_u\|. \quad (35)$$

This reveals that CFG systematically amplifies prediction errors as the guidance weight  $w$  increases, especially when branch predictions are imperfect.

**Multi-step accumulation and stage dependence.** Since diffusion sampling is iterative, amplified errors propagate and accumulate across timesteps, progressively pushing samples away from the training distribution. Notably, early low-SNR stages contribute disproportionately to drift accumulation while offering limited semantic refinement, whereas later high-SNR stages are more semantically informative but remain sensitive to magnitude distortion introduced by guidance.

**Implications for sampling design.** The above analysis indicates that uniformly allocating sampling steps across all timesteps is suboptimal. More importantly, the above analysis not only highlights the need for distribution expansion, but also reveals that the effectiveness of diffusion-based augmentation critically depends on generation stability. Specifically, prediction drift can potentially lead to unreliable teacher supervision on generated samples, while early low-SNR stages introduce excessive uncertainty that does not contribute to meaningful semantic refinement. Even when employing skip-step samplers such as DDIM, early low-SNR stages still consume a non-negligible portion of the sampling budget while introducing a heightened risk of prediction drift.

This issue is particularly critical in the source-domain-unknown setting, where no real source data are available to verify or correct the generated samples. As a result, unstable generation may directly introduce misleading teacher supervision signals, making generation stability essential for reliable knowledge distillation. The above analysis indicates that uniformly allocating sampling steps across all timesteps is suboptimal. These observations motivate two complementary design principles: (i) correcting the magnitude bias induced by guided noise prediction to suppress drift amplification, and (ii) shifting the sampling start point to later, higher-SNR timesteps to avoid error-prone early stages. In the following subsections, we formalize these principles through the proposed

DANR mechanism and the SSS strategy, which together constitute the core of our drift-corrected diffusion framework. Specifically, DANR is designed to suppress the error amplification induced by classifier-free guidance, while SSS mitigates the instability of early low-SNR timesteps by shifting the sampling process to more reliable stages. These observations further suggest that, unlike conventional diffusion designs primarily focused on perceptual fidelity or sampling efficiency, our objective is to explicitly control generation stability for reliable knowledge transfer. Accordingly, in our framework, the quality of generated samples is evaluated not only by perceptual realism, but more importantly by the reliability of the resulting distillation supervision signals.

Collectively, these principles aim to improve the stability and reliability of the generation process, which is critical for effective diffusion-based data augmentation in knowledge distillation.

#### 4.3. Overall framework of $D^3$ -KD

Motivated by the above theoretical analysis on distribution expansion and generation stability, we propose the  $D^3$ -KD framework for source-domain-unknown knowledge distillation, which incorporates diffusion-based data augmentation to improve domain alignment. Our contribution is not to propose entirely new diffusion mechanisms, but to reinterpret and redesign diffusion sampling strategies from the perspective of knowledge distillation, establishing a principled connection between distribution expansion and generation stability. Rather than introducing a new theoretical bound, we aim to identify how distribution expansion and generation stability jointly determine the effectiveness of knowledge distillation in this setting. As illustrated in Fig. 2(a), we perform source-to-target knowledge distillation by jointly leveraging input perturbation and diffusion-based data augmentation to improve the robustness and domain alignment of the student model. A composite distillation objective, combining perturbation-stabilized teacher supervision (Hossain et al., 2025) and relation-based feature alignment, is adopted as the default (Tung & Mori, 2019) KD instantiation and does not constitute the main contribution of this work.

**Perturbed teacher inference.** Given a target-domain sample  $(\mathbf{x}, \mathbf{y})$ , we first generate a perturbed input to stabilize the teacher's predictions:

$$\tilde{\mathbf{x}}_p = \mathcal{P}(\mathbf{x}), \quad (36)$$

where  $\mathcal{P}(\cdot)$  denotes a generic perturbation operator. The frozen teacher produces soft predictions:

$$p_T = \sigma(f_T(\mathbf{x})), \quad p_{T,p} = \sigma(f_T(\tilde{\mathbf{x}}_p)), \quad (37)$$

and the consistency between  $p_T$  and  $p_{T,p}$  is enforced to obtain more reliable supervisory signals.

**Diffusion-based data augmentation.** In parallel, we employ a DCED framework to further enrich the target-domain training data and reduce the distribution gap to the source domain:

$$\tilde{\mathbf{x}}_d = \mathcal{G}_{\text{DCED}}(\mathbf{x}), \quad (38)$$

where  $\tilde{\mathbf{x}}_d$  denotes an image generated from  $\mathbf{x}$ . Although diffusion models exhibit strong generative capability, their prediction bias and the inefficiency of multi-step sampling can lead to quality degradation and excessive computational cost. To address these limitations, we introduce DCED as an improved diffusion-based knowledge bridge, which enables efficient and stable generation for data augmentation. The detailed design of DCED is presented in Sections 4.4 and 4.6.

This diffusion-driven augmentation increases the diversity of the target-domain samples and facilitates better alignment with the source-domain teacher. The student predictions on the augmented data are:

$$p_d = \sigma(f_S(\tilde{\mathbf{x}}_d)), \quad (39)$$

while the clean target input is predicted as:

$$p = \sigma(f_S(\mathbf{x})), \quad (40)$$

where  $f_S(\cdot)$  denotes the student segmentation network and  $\sigma(\cdot)$  represents the pixel-wise activation function.

**Relational knowledge distillation.** To transfer high-level knowledge from the source-domain teacher, we employ a relation module that aligns intermediate representations between teacher and student, including feature alignment and prototype-based relational learning. The detailed formulation is given in subsequent subsections.

**Training objective.** As illustrated in Fig. 2(b), the student network is optimized under three complementary supervision signals: task-level segmentation supervision, teacher-guided logit distillation with perturbation stabilization, and relation-based knowledge distillation. The overall objective is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}}(p, \mathbf{y}) + \mathcal{L}_{\text{logit}}(p, p_T) + \mathcal{L}_{\text{cons}}(p_T, p_{T,p}) + \mathcal{L}_{\text{feat}} + \mathcal{L}_{\text{proto}}. \quad (41)$$

Here,  $\mathcal{L}_{\text{seg}}$  denotes the segmentation loss with ground-truth annotations,  $\mathcal{L}_{\text{logit}}$  enforces alignment between student predictions and teacher soft outputs,  $\mathcal{L}_{\text{cons}}$  stabilizes teacher supervision under input perturbations, and  $\mathcal{L}_{\text{feat}}$  and  $\mathcal{L}_{\text{proto}}$  correspond to feature- and prototype-level relational distillation, respectively.

#### 4.4. Structural details of the diffusion denoising network

The denoising network follows a U-Net-style encoder-decoder architecture with a bottleneck (Ronneberger et al., 2015; Yu et al., 2023). Given a noisy latent sample  $\mathbf{x}_{t_k}$  at diffusion step  $t_k$ , the network predicts the corresponding noise under optional mask-based conditioning.

##### 4.4.1. Encoder

The encoder progressively extracts hierarchical representations from the input  $\mathbf{x}_{t_k}$  by stacking multiple residual blocks. Within each resolution stage, feature refinement is performed through group normalization, nonlinear activation, and point-wise convolution. After all residual blocks at a given resolution are applied, a self-attention layer with skip connection (Wang et al., 2018) is employed to capture long-range dependencies, followed by a downsampling operation  $\text{Down}(\cdot)$  to reduce the spatial resolution:

$$\mathbf{z}_{t_k}^{(l)} = \mathcal{F}^{(2)}\left(\text{Inject}\left(\mathcal{F}^{(1)}(\mathbf{z}_{t_k}^{(l-1)}), t\right)\right), l = 1, \dots, L, \quad (42)$$

$$\mathbf{h}_{t_k} = \text{Down}\left(\text{Attn}(\mathbf{z}_{t_k}^{(L)})\right). \quad (43)$$

$$\mathcal{F}^{(l)}(\mathbf{x}) = \text{EncBlock}(\mathbf{x}), \quad (44)$$

where  $\text{EncBlock}^{(l)}(\cdot)$  denotes an *encoder module* that performs feature encoding through normalization, nonlinearity, and point-wise convolution.  $\mathbf{h}_{t_k}^{(0)} = \mathbf{x}_{t_k}$  and  $l$  denotes the encoder stage index. Temporal information is injected via a time embedding  $t_k$ , which is added to the intermediate features after normalization. After the first Group Normalization, SiLU activation, and  $1 \times 1$  convolution, temporal information is incorporated into the network via a feature-wise affine transformation. Specifically, learnable time-dependent scaling and shifting parameters  $\mathbf{w}(t) \in \mathbb{R}^{1 \times 1 \times C}$  and  $\mathbf{b}(t) \in \mathbb{R}^{1 \times 1 \times C}$  are applied to linearly modulate the intermediate activations, thereby injecting the diffusion timestep into the network.

##### 4.4.2. Bottleneck

The deepest encoder features are processed by a bottleneck block to capture global context:

$$\mathbf{h}_{t_k}^{\text{bot}} = \text{Bottleneck}(\mathbf{h}_{t_k}^{(L)}), \quad (45)$$

where  $L$  is the total number of encoder stages.

##### 4.4.3. Decoder with conditional guidance

In the decoder, the upsampled feature  $\tilde{\mathbf{h}}_{t_k}^{(l)}$  is first refined through a stack of SPADE-conditioned convolutional blocks (Park et al., 2019), where structural guidance  $\mathbf{y}_{t_{k-1}}$  is progressively injected. After all decoder blocks at the current resolution are applied, a self-attention layer

is employed to capture global dependencies, followed by an upsampling operation  $\text{Up}(\cdot)$  to recover spatial resolution.

$$\mathbf{z}_{t_k}^{(l)} = \mathcal{G}^{(2)}\left(\text{Inject}\left(\mathcal{G}^{(1)}(\mathbf{z}_{t_k}^{(l)}, \mathbf{y}_{t_{k-1}}), t\right), \mathbf{y}_{t_{k-1}}\right), \quad (46)$$

$$l = 1, \dots, L.$$

$$\mathbf{h}_{t_k}^{(l)} = \text{Up}\left(\text{Attn}(\mathbf{z}_{t_k}^{(L)})\right). \quad (47)$$

$$\mathcal{G}^{(i)}(\mathbf{x}, \mathbf{y}) = \text{Decoder}^{(i)}(\mathbf{x}, \mathbf{y}), \quad i \in \{1, 2\}, \quad (48)$$

where  $\mathbf{z}_{t_k}^{(0)} = \tilde{\mathbf{h}}_{t_k}^{(l)}$ , and  $\text{Decoder}^{(i)}(\cdot)$  denotes a mask-guided conditional decoding module.  $\mathbf{y}_{t_{k-1}}$  is obtained via mask augmentation and geometric transformation:

$$\mathbf{y}_{t_{k-1}} = \mathcal{T}(\text{MaskAug}(\mathbf{y})). \quad (49)$$

where  $\text{MaskAug}(\cdot)$  denotes the mask augmentation operation. Self-attention layers are additionally inserted at selected decoder resolutions to enhance long-range dependency modeling. Finally, the decoder outputs the predicted noise:

$$\epsilon_{\theta}(\mathbf{x}_{t_k}, t_k, \mathbf{y}_{t_{k-1}}) = \text{Conv}_{1 \times 1}(\mathbf{h}_{t_k}^{(0)}), \quad (50)$$

which is used to compute the next latent state  $\mathbf{x}_{t_{k-1}}$  through the reparameterization.

#### 4.5. Structure-aware diffusion training

To ensure anatomically plausible lesion synthesis, the diffusion model is trained with structure-aware supervision that incorporates semantic priors during denoising. Instead of relying solely on pixel-wise noise reconstruction, we introduce auxiliary structural constraints to guide the generative process toward lesion-relevant regions while suppressing background interference. Specifically, the overall training objective is defined as:

$$\mathcal{L}_{\text{diff}} = \mathcal{L}_{\text{noise}}^{\text{ada}} + \mathcal{L}_{\text{refine}}, \quad (51)$$

where  $\mathcal{L}_{\text{noise}}^{\text{ada}}$  is a mask-adaptive noise prediction loss that emphasizes lesion regions during denoising, and  $\mathcal{L}_{\text{refine}}$  denotes a lightweight structural refinement regularization using a pre-trained and frozen segmentation network.

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#### Algorithm 1 Sampling process.

---

**Require:** Denoiser  $\epsilon_{\theta}(\cdot)$ ; conditioning  $\mathbf{y}$ ; guidance scale  $\lambda$ ; rescale strength  $\rho$ ; timestep schedule  $\{t_0, t_1, \dots, t_K\}$  with  $t_0 > t_1 > \dots > t_K = 0$ ; cumulative noise schedule  $\{\bar{\alpha}_t\}$ .

**Ensure:** Generated sample  $\mathbf{x}_0$ .

```

1: Sample  $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ 
2: Initialize  $\mathbf{x}_{t_0} = \gamma \mathbf{z}$ 
3: for  $k = 0$  to  $K - 1$  do
4:    $t \leftarrow t_k, \quad t' \leftarrow t_{k+1}$                                 {next timestep}
5:    $\epsilon_c \leftarrow \epsilon_{\theta}(\mathbf{x}_t, t)$                                     {conditional}
6:    $\epsilon_u \leftarrow \epsilon_{\theta}(\mathbf{x}_t, t)$                                     {unconditional}
7:    $\epsilon_{\text{cfg}} \leftarrow (1 + \lambda)\epsilon_c - \lambda\epsilon_u$                         {Eq. (52)}
8:    $s \leftarrow \sigma(\epsilon_u) / \sigma(\epsilon_{\text{cfg}})$                         { $\sigma(\cdot)$  over spatial + channel}
9:    $\epsilon_{\text{res}} \leftarrow \epsilon_u + \rho \cdot s \cdot (\epsilon_{\text{cfg}} - \epsilon_u)$             {Eq. (54)}
10:   $\hat{\mathbf{x}}_0 \leftarrow \frac{1}{\sqrt{\bar{\alpha}_t}} \left( \mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_{\text{res}} \right)$     {Eq. (56)}
11:   $\sigma_t \leftarrow \sqrt{\frac{1 - \bar{\alpha}_{t'}}{1 - \bar{\alpha}_t} \left( 1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t'}} \right)}$ 
12:   $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ 
13:   $\mathbf{x}_{t'} \leftarrow \sqrt{\bar{\alpha}_{t'}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'} - \sigma_t^2} \epsilon_{\text{rescaled}} + \sigma_t \mathbf{z}$ 
14: end for
15: return  $\mathbf{x}_0 \leftarrow \mathbf{x}_{t_K}$ 
```

---

#### 4.6. Drift-corrected efficient diffusion

To enable efficient and stable generation under aggressive timestep skipping, we adopt a DDIM-style sampling framework (Song et al., 2021) and incorporate CFG, drift-aware noise rescaling, and jump-efficient sampling. The complete procedure is summarized in Algorithm 1.

##### 4.6.1. Overall sampling process

Given a predefined timestep sequence  $\{t_0, t_1, \dots, t_K\}$  with  $t_0 > t_1 > \dots > t_K = 0$ , we skip early timesteps to shorten the sampling trajectory and initialize  $\mathbf{x}_{t_0}$  from a standard Gaussian distribution. At each timestep  $t_k$ , the denoising network predicts the noise component corresponding to the current latent state.

To enhance conditional generation, CFG (Ho & Salimans, 2021) is applied by jointly considering conditional and unconditional noise predictions,  $\epsilon_\theta(\mathbf{x}_{t_k}, t_k, \mathbf{y})$  and  $\epsilon_\theta(\mathbf{x}_{t_k}, t_k)$ , respectively. Both predictions are produced by the same denoising network, where the unconditional branch is obtained by randomly dropping the conditioning variable during training. The guided noise prediction is computed as:

$$\epsilon_{\text{cfg}} = (1 + \lambda) \epsilon_\theta(\mathbf{x}_{t_k}, t, \mathbf{y}) - \lambda \epsilon_\theta(\mathbf{x}_{t_k}, t), \quad (52)$$

where  $\lambda$  controls the strength of conditional guidance.

##### 4.6.2. Drift-aware noise reconciliation

This design is directly motivated by the drift amplification analysis in Section 4.2. CFG enhances conditional fidelity by extrapolating between conditional and unconditional noise predictions. However, as analyzed in Section 4.2, this extrapolation also amplifies prediction errors, particularly under skip-step sampling, leading to severe drift accumulation. A naive solution that uniformly rescales the guided noise or adjusts the guidance weight fails to address this issue, as it indiscriminately alters both semantic direction and noise magnitude.

While existing classifier-free guidance calibration or noise rescaling strategies primarily focus on improving visual fidelity or sampling quality, the proposed DANR explicitly targets the suppression of drift accumulation under skip-step diffusion sampling for knowledge distillation, where stable and reliable teacher supervision is critical.

To explicitly suppress guidance-induced drift while preserving conditional semantics, we propose a drift-aware noise reconciliation mechanism that rebalances the magnitude of the guided noise using the unconditional branch as a statistical reference. Formally, the reconciled noise prediction at timestep  $t_k$  is defined as:

$$s_{t_k} = \frac{\sigma(\epsilon_\theta(\mathbf{x}_{t_k}, t_k))}{\sigma(\epsilon_{\text{cfg}})}, \quad (53)$$

$$\epsilon_{\text{rescaled}} = \epsilon_\theta(\mathbf{x}_{t_k}, t_k) + \rho s_{t_k} (\epsilon_{\text{cfg}} - \epsilon_\theta(\mathbf{x}_{t_k}, t_k)), \quad (54)$$

where  $\sigma(\epsilon_\theta)/\sigma(\epsilon_{\text{cfg}})$  acts as a variance-alignment factor to calibrate the magnitude of the guided noise.  $\epsilon_\theta(\mathbf{x}_{t_k}, t_k)$  denotes the unconditional noise prediction,  $\epsilon_{\text{cfg}}$  is the classifier-free guided noise, and  $\sigma(\cdot)$  computes the standard deviation over spatial and channel dimensions. The coefficient  $\rho \in [0, 1]$  controls the strength of noise calibration.

Unlike naive noise scaling, Eq. (54) operates on the *incremental guidance term* ( $\epsilon_{\text{cfg}} - \epsilon_\theta$ ). This design preserves the conditional direction introduced by classifier-free guidance, while explicitly regulating its magnitude using the noise statistics of the unconditional branch. Since  $\epsilon_\theta$  reflects the noise scale encountered during training, the ratio  $\sigma(\epsilon_\theta)/\sigma(\epsilon_{\text{cfg}})$  re-aligns the guided prediction to a more stable and training-consistent scale.

From the perspective of prediction drift (Section 4.2), drift accumulation arises because the amplified noise under CFG is recursively propagated across denoising steps. By constraining the effective noise deviation at each timestep, DANR reduces the per-step error amplification rate, thereby suppressing the accumulation of prediction drift over the entire sampling trajectory.

##### 4.6.3. Shifted-start sampling strategy

Unlike conventional diffusion sampling that starts from the terminal noise state ( $t = T$ ), the proposed SSS strategy initializes the reverse process from an intermediate timestep  $t_s < T$ .

While previous timestep truncation or fast sampling strategies focus on reducing computational cost, the proposed SSS is motivated by the observation that early low-SNR stages introduce disproportionate instability, and is designed to improve generation reliability for downstream knowledge distillation. This design explicitly avoids early low-SNR regions, where noise predictions contribute little to semantic refinement yet introduce disproportionately large prediction uncertainty and error amplification. In the SFDA-KD setting, such unstable predictions may further corrupt the teacher supervision signal and accumulate throughout student optimization, making early-stage diffusion errors particularly harmful for reliable knowledge transfer.

Rather than drawing  $\mathbf{x}_{t_s}$  from the exact forward diffusion process, we directly sample an initial latent state from a Gaussian distribution:

$$\mathbf{x}_{t_s} = \gamma \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (55)$$

where  $\gamma > 1$  is a scaling factor introduced to compensate for the variance mismatch induced by skipping the early diffusion steps. Since the omitted forward transitions would have gradually accumulated noise variance, directly sampling from  $\mathcal{N}(\mathbf{0}, \mathbf{I})$  would underestimate the marginal variance at timestep  $t_s$ . The scaling factor  $\gamma$  re-aligns the initialized latent state with the noise manifold of the true diffusion process, thereby preserving distributional consistency.

Given the rescaled noise prediction at timestep  $t_s$ , the corresponding clean sample is estimated as:

$$\hat{\mathbf{x}}_0 = \frac{1}{\sqrt{\bar{\alpha}_{t_s}}} (\mathbf{x}_{t_s} - \sqrt{1 - \bar{\alpha}_{t_s}} \epsilon_{\text{rescaled}}). \quad (56)$$

Starting from the shifted initialization, the latent state at the current timestep  $t$  is propagated to the next sampled timestep  $t' < t$  using a DDIM-style update:

$$\mathbf{x}_{t'} = \sqrt{\bar{\alpha}_{t'}} \hat{\mathbf{x}}_0 + \sqrt{1 - \bar{\alpha}_{t'} - \sigma_t^2} \epsilon_{\text{rescaled}} + \sigma_t \mathbf{z}, \quad (57)$$

where  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ .

To maintain consistency with the original diffusion dynamics under the shifted sampling schedule, the stochastic noise scale is defined as

$$\sigma_t = \sqrt{\frac{1 - \bar{\alpha}_{t'}}{1 - \bar{\alpha}_t} \left(1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t'}}\right)}, \quad t' < t. \quad (58)$$

By jointly shifting the sampling start and calibrating both the initial noise variance and subsequent stochasticity, SSS effectively shortens the sampling trajectory while suppressing the propagation of early-stage prediction errors. As a result, avoiding instability-dominated early diffusion stages, SSS improves the reliability of generated supervision signals and reduces the accumulation of prediction drift during knowledge distillation. Importantly, SSS complements the proposed DANR mechanism: while DANR constrains the magnitude of prediction errors at each denoising step, SSS reduces the total number of error-prone steps by avoiding unreliable early stages. Together, they form a unified strategy for mitigating prediction drift under accelerated diffusion sampling.

## 5. Experiments and analysis

In this section, we conduct extensive experiments to evaluate the proposed D<sup>3</sup>-KD framework under realistic source-domain-unknown clinical scenarios. We first introduce the experimental setup, including datasets, evaluation protocols, and implementation details. We then present comprehensive comparisons with representative knowledge distillation and related baselines to demonstrate the overall effectiveness and universality of our approach. Finally, we provide in-depth analyses of the diffusion-based generative bridge, including ablation studies and efficiency evaluations, to clarify how each component contributes to the observed performance gains.

**Table 2**

Model complexity comparison between teacher and student networks.

| Model                     | Params (M) | GFLOPs (G) |
|---------------------------|------------|------------|
| EfficientNet-B7 (Teacher) | 66.34      | 37.75      |
| EfficientNet-B0 (Student) | 5.28       | 0.39       |
| Swin-B (Teacher)          | 88.00      | 15.40      |
| MobileNetV2 (Student)     | 3.50       | 0.30       |
| BUSSAM (ViT-B, Teacher)   | 91.87      | 105.87     |
| ResUNetSE (Student)       | 17.59      | 30.79      |

## 5.1. Experimental setup

### 5.1.1. Datasets and experimental protocol

To evaluate the proposed source-domain-unknown knowledge distillation framework under realistic clinical conditions, we conduct experiments on multiple breast ultrasound datasets collected from heterogeneous acquisition settings, which naturally induce substantial domain discrepancies.

The BUS-BRA dataset (Gómez-Flores et al., 2024) is used for teacher model pre-training. It is a large-scale multi-center breast ultrasound dataset collected from multiple hospitals and ultrasound manufacturers, covering diverse imaging styles. Due to data privacy regulations and cross-institutional data-sharing restrictions, the source-domain data are assumed to be strictly inaccessible during student training, forming a realistic source-domain-unknown distillation setting.

The BUS-UCLM dataset (Vallez et al., 2025) serves as the target domain for student training and evaluation. It is a single-center dataset acquired using a Siemens S2000 ultrasound system, resulting in consistent acquisition protocols but a pronounced domain shift relative to BUS-BRA, reflecting typical clinical deployment conditions with limited target-domain data.

To mitigate the distribution gap between the unknown source domain and the target domain, we construct an auxiliary dataset, termed BUS-UCLM + +, using the proposed DCED framework. BUS-UCLM + + expands the support of the target-domain distribution by generating structurally guided and anatomically plausible ultrasound images, thereby increasing its potential overlap with the unknown source domain.

Unless otherwise specified, teacher models are pre-trained on BUS-BRA, while student models are trained on BUS-UCLM with the assistance of BUS-UCLM + +. No source-domain data are accessed during student training, ensuring a strict source-domain-unknown experimental protocol.

### 5.1.2. Implementation details

All experiments were implemented in PyTorch using Python and conducted on a high-performance workstation equipped with two NVIDIA GeForce RTX 4090 GPUs (24 GB memory each). CUDA 12.8 and cuDNN were utilized to accelerate both training and inference.

**Hyperparameter Settings.** The student network was optimized using the SGD optimizer with a momentum of 0.9 and a weight decay of  $1 \times 10^{-4}$ . The initial learning rate was set to 0.01 and decayed according to a poly policy with power 0.9. The batch size was set to 8, and the training epoch was set to 100. Regarding the distillation objective in Eq. (41), the loss balancing coefficients were empirically set as follows:  $\lambda_{\text{logit}} = 0.5$  for logit-level alignment,  $\lambda_{\text{feat}} = 0.05$  for feature-level distillation, and  $\lambda_{\text{proto}} = 0.02$  for prototype matching. The perturbation consistency weight was set to  $\lambda_{\text{cons}} = 0.01$  to ensure training stability. For the diffusion process, the noise rescaling factor was set to  $\rho = 0.9$  and the initialization scale  $\gamma = 1.00001$ , as determined by our ablation study.

### 5.1.3. Teacher and student network selection

To ensure a fair and representative evaluation of the proposed source-domain-unknown knowledge distillation framework, we carefully select teacher and student networks with distinct capacities and architectural characteristics.

For the teacher models, we employ high-capacity architectures that have demonstrated strong representation ability in both natural and medical imaging tasks. Specifically, EfficientNet-B7 (Tan & Le, 2019) and Swin-B (Liu et al., 2021b) are adopted as representative CNN- and Transformer-based teachers, respectively, due to their strong feature extraction capability and wide adoption in prior distillation studies. In addition, we use BUSSAM (Tu et al., 2024), a ViT-B-based medical-specific model, as a specialized teacher for breast ultrasound segmentation, reflecting a realistic scenario where large foundation-style models are available but impractical for deployment.

For the student models, we select lightweight networks that are suitable for clinical deployment under limited computational resources. EfficientNet-B0 and MobileNetV2 are chosen as compact CNN backbones with favorable accuracy–efficiency trade-offs, while ResUNetSE (Yu et al., 2022) is employed as a lightweight segmentation network commonly used in vision applications. These student architectures are designed to maintain reasonable representational capacity while significantly reducing computational and memory costs.

As summarized in Table 2, the selected teacher models contain between 66.3M and 91.9M parameters, with computational complexity ranging from 15.4 to 105.9 GFLOPs. In contrast, the student models require only 3.5M–17.6M parameters and substantially fewer GFLOPs, highlighting the practical importance of effective knowledge distillation for bridging this large capacity gap.

Unless otherwise specified, BUSSAM and ResUNetSE are adopted as the default teacher–student pair for all subsequent ablation studies and hyperparameter analyses, providing a consistent and controlled experimental setting.

### 5.1.4. Evaluation metrics

We evaluate the segmentation performance using Dice ( $\uparrow$ ), Intersection over Union (IoU,  $\uparrow$ ), Recall ( $\uparrow$ ), and the 95th percentile Hausdorff Distance (HD95,  $\downarrow$ ). Let  $P$  denote the predicted segmentation mask and  $G$  the ground-truth mask.

**Dice Coefficient.** Dice measures the overlap between the predicted region and the ground truth:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}. \quad (59)$$

**Intersection over Union (IoU).** IoU evaluates the region-level overlap ratio:

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|}. \quad (60)$$

**Recall.** Recall reflects the sensitivity of lesion detection:

$$\text{Recall} = \frac{|P \cap G|}{|G|}. \quad (61)$$

**95th Percentile Hausdorff Distance (HD95).** HD95 measures boundary accuracy by computing the 95th percentile of the Hausdorff distance between the prediction boundary  $\partial P$  and the ground-truth boundary  $\partial G$ :

$$\text{HD95} = \max \left\{ \text{percentile}_{95} (d(\partial P, \partial G)), \text{percentile}_{95} (d(\partial G, \partial P)) \right\}, \quad (62)$$

where  $d(\cdot, \cdot)$  denotes the Euclidean distance.

**Table 3**

Quantitative comparison with state-of-the-art adaptive methods on the BUS-UCLM dataset. Best results are highlighted in **bold**, and second-best results are underlined. Metrics include dice, IoU, and recall (%), and HD95 (pixels).

| Adaptive Method                | Teacher → Student Architecture             | Metrics on BUS-UCLM (Target Domain) |              |              |              |
|--------------------------------|--------------------------------------------|-------------------------------------|--------------|--------------|--------------|
|                                |                                            | Dice (↑)                            | IoU (↑)      | Recall (↑)   | HD95 (↓)     |
| Source-Only                    |                                            | 56.99                               | 39.85        | 91.95        | 16.72        |
| AdaBN (Li et al., 2018)        |                                            | 67.69                               | 51.16        | 81.52        | 17.50        |
| SHOT (Liang et al., 2020)      |                                            | 67.54                               | 50.99        | 82.21        | 16.52        |
| TENT (Wang et al., 2021)       |                                            | 67.53                               | 50.98        | 80.00        | 25.79        |
| COTTA (Wang et al., 2022)      |                                            | 67.65                               | 51.12        | 81.09        | 25.07        |
| SAR (Niu et al., 2023)         | <i>EfficientNet-B7 → EfficientNet-B0</i>   | 65.76                               | 48.99        | 81.37        | 11.92        |
| D-master (Ashraf et al., 2024) |                                            | 69.20                               | 52.10        | 81.50        | 13.50        |
| VPTTA (Chen et al., 2024)      |                                            | <u>71.80</u>                        | <u>54.90</u> | <u>82.50</u> | <u>11.50</u> |
| UCON (Xu et al., 2025a)        |                                            | 67.81                               | 51.30        | 80.99        | 13.72        |
| LNR (Xu et al., 2025b)         |                                            | 67.71                               | 51.18        | 81.47        | 14.95        |
| AIF-SFDA (Li et al., 2025)     |                                            | 70.50                               | 53.80        | 82.10        | 12.20        |
| <b>Ours (D<sup>3</sup>-KD)</b> |                                            | <b>94.60</b>                        | <b>89.75</b> | <b>95.03</b> | <b>3.11</b>  |
| Source-Only                    |                                            | 74.78                               | 59.72        | 85.06        | 18.35        |
| AdaBN (Li et al., 2018)        |                                            | 79.50                               | 65.98        | 85.32        | 9.40         |
| SHOT (Liang et al., 2020)      |                                            | 79.62                               | 66.15        | 85.18        | 12.55        |
| TENT (Wang et al., 2021)       |                                            | 79.95                               | 66.60        | 85.02        | 14.39        |
| COTTA (Wang et al., 2022)      |                                            | 79.50                               | 65.98        | 85.28        | 8.12         |
| SAR (Niu et al., 2023)         | <i>Swin-Transformer-Base → MobileNetV2</i> | 78.12                               | 64.09        | 84.99        | 9.23         |
| D-master (Ashraf et al., 2024) |                                            | 78.45                               | 64.20        | 85.12        | 16.20        |
| VPTTA (Chen et al., 2024)      |                                            | <u>81.35</u>                        | <u>68.20</u> | <u>86.15</u> | 8.15         |
| UCON (Xu et al., 2025a)        |                                            | 79.01                               | 65.30        | 85.09        | 6.24         |
| LNR (Xu et al., 2025b)         |                                            | 79.40                               | 65.84        | 84.76        | 11.05        |
| AIF-SFDA (Li et al., 2025)     |                                            | 80.24                               | 67.12        | 85.80        | 12.50        |
| <b>Ours (D<sup>3</sup>-KD)</b> |                                            | <b>92.09</b>                        | <b>85.34</b> | <b>94.45</b> | <b>2.32</b>  |
| Source-Only                    |                                            | 44.98                               | 29.01        | 43.50        | 18.25        |
| AdaBN (Li et al., 2018)        |                                            | 58.69                               | 41.53        | 64.61        | 10.13        |
| SHOT (Liang et al., 2020)      |                                            | 50.90                               | 34.03        | 51.26        | 17.47        |
| TENT (Wang et al., 2021)       |                                            | 54.13                               | 37.11        | 59.49        | 14.23        |
| COTTA (Wang et al., 2022)      |                                            | 56.05                               | 38.94        | 61.45        | 19.84        |
| SAR (Niu et al., 2023)         | <i>BUSSAM → ResUNetSE</i>                  | 53.70                               | 36.71        | 65.63        | 13.25        |
| D-master (Ashraf et al., 2024) |                                            | 60.15                               | 43.20        | 65.10        | 9.80         |
| VPTTA (Chen et al., 2024)      |                                            | <u>62.80</u>                        | <u>45.80</u> | <u>67.50</u> | <u>8.10</u>  |
| UCON (Xu et al., 2025a)        |                                            | 57.58                               | 40.43        | 52.95        | 12.62        |
| LNR (Xu et al., 2025b)         |                                            | 57.50                               | 40.35        | 52.90        | 13.20        |
| AIF-SFDA (Li et al., 2025)     |                                            | 61.40                               | 44.50        | 66.30        | 8.90         |
| <b>Ours (D<sup>3</sup>-KD)</b> |                                            | <b>91.63</b>                        | <b>84.55</b> | <b>92.10</b> | <b>2.05</b>  |

## 5.2. Quantitative comparison with state-of-the-art methods

Table 3 presents a comprehensive quantitative comparison between the proposed D<sup>3</sup>-KD framework and a diverse set of representative methods on the BUS-UCLM target domain under different teacher–student configurations.

To ensure a fair and comprehensive evaluation, we include three categories of baselines: (i) test-time adaptation (TTA) methods (e.g., TENT (Wang et al., 2021), CoTTA (Wang et al., 2022), SAR (Niu et al., 2023), UCON (Xu et al., 2025a), LNR (Xu et al., 2025b), and VPTTA (Chen et al., 2024)), (ii) source-free domain adaptation (SFDA) methods for medical imaging (e.g., AIF-SFDA (Li et al., 2025)), and (iii) domain adaptation methods for breast imaging (e.g., D-MASTER (Ashraf et al., 2024)), which, although not strictly source-free, provide task-relevant comparisons. Although some baselines (e.g., SAR, UCON, and LNR) are originally developed for general test-time adaptation rather than breast ultrasound segmentation, they are included due to their shared constraint of adapting models without access to source-domain data, making them relevant for evaluating robustness under source-domain-unknown conditions.

Generally, existing methods yield only moderate improvements over the Source-Only baseline and remain sensitive to severe domain shifts. Even the strongest baselines, such as VPTTA and AIF-SFDA, achieve limited gains, indicating that adapting models without explicit access to source-domain data remains highly challenging. In contrast, D<sup>3</sup>-KD consistently delivers substantial performance improvements across all

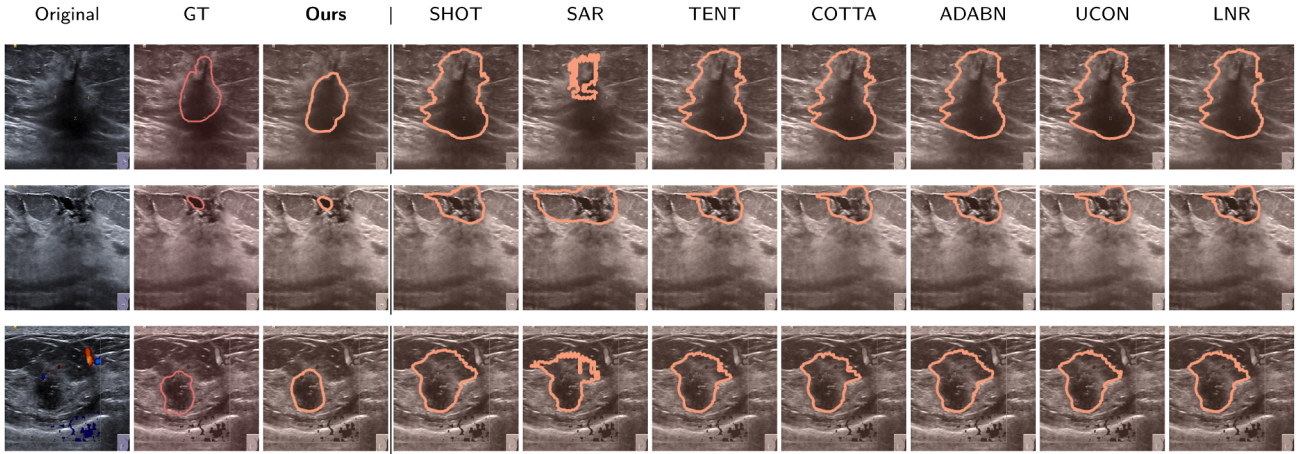
metrics, demonstrating the effectiveness of diffusion-enhanced knowledge distillation.

In the homogeneous CNN setting (EfficientNet-B7 → EfficientNet-B0), the Source-Only model suffers from severe over-segmentation, as reflected by its low Dice score and large HD95. While existing methods improve region-level metrics to some extent, they fail to accurately capture structural boundaries. By contrast, D<sup>3</sup>-KD significantly boosts the Dice score to 94.60% and reduces HD95 to 3.11, indicating superior structural preservation and boundary coherence.

For the heterogeneous Transformer-to-CNN setting (Swin-Transformer-Base → MobileNetV2), the domain gap is further exacerbated by architectural mismatch. Most baselines achieve Dice scores below 82%, even with advanced adaptation strategies. In comparison, D<sup>3</sup>-KD enables the lightweight student model to reach 92.09% Dice with an HD95 of 2.32, highlighting its strong generalization capability across heterogeneous architectures.

In the most challenging BUSSAM → ResUNetSE setting, the Source-Only baseline collapses to 44.98% Dice, and existing methods provide only limited recovery. Even the best-performing baseline (VPTTA) remains significantly behind. Remarkably, D<sup>3</sup>-KD elevates the Dice score to 91.63% while sharply reducing HD95 to 2.05, demonstrating its robustness under extreme domain shifts.

Overall, these results suggest that while TTA and SFDA methods can partially improve model robustness, they remain insufficient for reliable student learning under source-domain-unknown conditions. This is mainly because most existing approaches focus on inference-time model



**Fig. 3.** Qualitative comparison of breast cancer segmentation results on the BUS-UCLM dataset. Each row demonstrates a challenging breast ultrasound sample with high speckle noise. From left to right: the original input image, ground truth (GT), our proposed D<sup>3</sup>-KD method, and results from seven state-of-the-art SFDA and TTA methods. Our method exhibits superior robustness in boundary delineation and effectively suppresses domain-induced artifacts compared to baseline competitors.

adaptation rather than improving the reliability of transferable supervision signals. In contrast, D<sup>3</sup>-KD establishes a more effective knowledge transfer paradigm by explicitly expanding the target-domain support through diffusion-based data generation, leading to accurate and structurally consistent segmentation.

### 5.3. Qualitative visualization

To qualitatively assess the robustness of the proposed framework under severe domain shifts, Fig. 3 presents visual comparisons of breast lesion segmentation results on the BUS-UCLM target domain. Three representative cases are selected to illustrate common challenges in breast ultrasound imaging, including low lesion contrast (Row 1), complex background textures (Row 2), and acoustic shadowing artifacts (Row 3).

As shown in the figure, existing source-free domain adaptation (SFDA) and test-time adaptation (TTA) methods exhibit limited robustness under these challenging conditions. In particular, methods such as SHOT and SAR (Cols. 4–5) are highly sensitive to speckle noise, often producing fragmented predictions or missing lesion regions entirely (e.g., Row 1). While TENT and COTTA (Cols. 6–7) improve overall lesion coverage, they frequently over-segment surrounding tissues, resulting in irregular boundaries and inaccurate shape delineation. Even the strongest baseline, AdaBN (Col. 8), although capable of roughly localizing the lesion, still suffers from jagged contours and isolated false-positive regions.

In contrast, the proposed D<sup>3</sup>-KD framework (Col. 3) consistently generates smooth and topologically coherent segmentation masks that closely align with the ground truth annotations (Col. 2). By leveraging diffusion-driven data expansion and structure-aware knowledge transfer, our method effectively suppresses domain-induced artifacts and accurately distinguishes lesion boundaries from acoustic shadows and background clutter. These qualitative results further corroborate the quantitative findings and highlight the superior robustness and clinical applicability of the proposed approach under source-domain-unknown conditions.

### 5.4. Structural fidelity and generative quality analysis

As visually compared in Fig. 4, we evaluate the generative quality of the proposed framework against standard generative baselines, including DG-GAN (Liu et al., 2021a) and VAE (Kingma & Welling, 2014). While DG-GAN (Fig. 4g) tends to suffer from mode collapse or generate high-frequency artifacts, and VAE (Fig. 4h) often produces overly smoothed textures with blurred boundaries, our mask-guided diffusion model (Fig. 4b/e) generates anatomically plausible ultrasound images

with high-fidelity speckle noise. Crucially, the overlay visualizations (Fig. 4c/f) demonstrate that our generated lesions strictly adhere to the input masks, ensuring the structural consistency and semantic correctness required for reliable pixel-level supervision.

Beyond visual augmentation, these high-quality generated samples serve as a pivotal *distributional bridge* between the target domain and the inaccessible source domain. By injecting structurally controlled yet stylistically diverse generated images into the training set, we effectively expand the support of the target-domain data manifold. This expansion increases the likelihood of the student network encountering features that overlap with the unknown source-domain distribution learned by the teacher, thereby indirectly reducing the domain gap. Consequently, the student model can better absorb the teacher's transferred knowledge without requiring direct access to the source data.

#### 5.4.1. Sensitivity analysis of synthetic data ratio

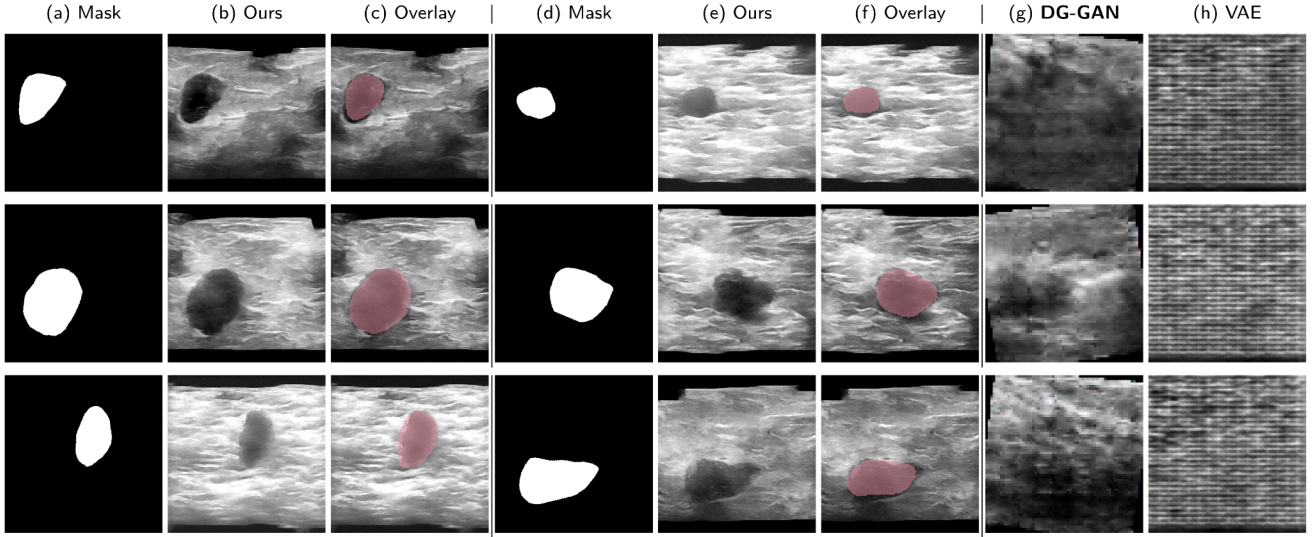
We further analyze the sensitivity of the proposed framework to the volume of synthetic samples introduced during training. Let  $\lambda \in [0, 1]$  denote the augmentation ratio of synthetic data relative to the original real dataset. In this setting,  $\lambda = 0$  corresponds to training with only real samples, while  $\lambda = 1$  indicates expanding the dataset with an equal number of synthetic samples (i.e., a 1:1 real-to-synthetic ratio), resulting in a distribution where synthetic data constitutes 50% of the total training set.

As illustrated in Fig. 5, increasing the synthetic data ratio consistently improves segmentation performance across multiple metrics. In particular, both Dice and mIoU scores steadily increase with  $\lambda$  and reach their best performance at  $\lambda = 1$ , corresponding to a 1:1 ratio of synthetic to real samples. This indicates that high-quality synthetic images effectively enrich the training distribution and provide complementary supervision while maintaining stable training behavior. Boundary accuracy exhibits an even more pronounced improvement. As shown by the HD95 curve, the Hausdorff distance is substantially reduced from 6.05 to a minimum of 2.05 as the synthetic ratio increases, suggesting that the mask-guided diffusion process provides reliable structural cues and improves the model's ability to capture complex lesion boundaries.

Overall, these results suggest that a balanced integration of synthetic and real samples yields the most effective knowledge transfer.

### 5.5. Universality of diffusion augmentation and method analysis

Table 4 evaluates the effect of DCED-generated data across a broader range of knowledge distillation methods, including standard KD (Hinton et al., 2015), FitNet (Romero et al., 2015), KD + Consistency (Tarvainen & Valpola, 2017), SKD (Liu et al., 2019), IFVD (Wang et al., 2020),

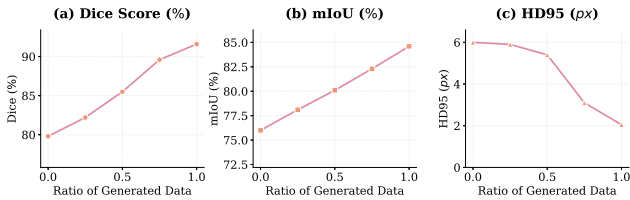


**Fig. 4.** Visual fidelity and generative quality assessment. Cols (a-f) showcase our method’s ability to generate diverse, high-fidelity ultrasound images that strictly adhere to the input lesion masks (overlay). Cols (g-h) display representative samples from DG-GAN and VAE baselines for texture comparison. While DG-GAN (g) often suffers from mode collapse and artifacts, and VAE (h) tends to produce blurry outputs, our method achieves superior textural realism and structural definition.

**Table 4**

Universality of diffusion-based augmentation. We compare different distillation methods with and without diffusion-generated data. The “real only” column aligns with our preliminary analysis (Table 1), where student only achieves 79.56% dice and standard KD reaches 81.25%. Incorporating DCED-generated data consistently improves all methods, while our  $D^3$ -KD achieves the best performance of 91.63% Dice.

| Method                                       | Setting A: Real Only |                   | Setting B: Real + DCED |                   | Gain ( $\Delta$ ) |
|----------------------------------------------|----------------------|-------------------|------------------------|-------------------|-------------------|
|                                              | Dice (%) $\uparrow$  | HD95 $\downarrow$ | Dice (%) $\uparrow$    | HD95 $\downarrow$ |                   |
| Student Only                                 | 79.56                | 4.12              | 83.50                  | 3.65              | +3.94             |
| Standard KD (Hinton et al., 2015)            | 81.25                | 3.85              | 88.30                  | 2.95              | +7.05             |
| FitNet (Romero et al., 2015)                 | 83.14                | 3.42              | 89.45                  | 2.65              | +6.31             |
| KD + Consistency (Tarvainen & Valpola, 2017) | 84.05                | 3.10              | 90.15                  | 2.38              | +6.10             |
| SKD (Liu et al., 2019)                       | 85.23                | 3.07              | 89.40                  | 2.58              | +4.17             |
| IFVD (Wang et al., 2020)                     | 84.51                | 3.12              | 90.77                  | 2.39              | +6.26             |
| CWD (Shu et al., 2021)                       | 85.30                | 3.01              | 91.37                  | 2.35              | +6.07             |
| LSCD (Wang et al., 2023a)                    | 84.74                | 3.15              | 89.87                  | 2.95              | +5.13             |
| PRRD (Wang et al., 2023b)                    | 84.09                | 3.85              | 90.97                  | 2.18              | +6.88             |
| RSKD (Liang et al., 2024)                    | 84.46                | 3.02              | 90.29                  | 2.21              | +5.83             |
| <b><math>D^3</math>-KD (Ours)</b>            | <b>85.40</b>         | <b>2.98</b>       | <b>91.63</b>           | <b>2.05</b>       | <b>+6.23</b>      |



**Fig. 5.** Effect of the synthetic data ratio  $\lambda$  on knowledge distillation performance. The mixed training distribution is defined as  $P'_T = \frac{1}{1+\lambda} P_T + \frac{\lambda}{1+\lambda} P_G$ , where  $\lambda$  denotes the ratio of generated to real target samples.  $\lambda = 0$  indicates real-only training, whereas  $\lambda = 1$  indicates a 1:1 real-to-generated mixture.

CWD (Shu et al., 2021), LSCD (Wang et al., 2023a), PRRD (Wang et al., 2023b), and RSKD (Liang et al., 2024), to provide a comprehensive comparison under the source-domain-unknown setting.

Under the *Real Only* setting, Student Only and Standard KD obtain 79.56% and 81.25% Dice, respectively, consistent with the preliminary results in Table 1. The stronger distillation baselines improve the Dice score to approximately 84–85%, but their performance remains limited when only target-domain samples are available. After adding DCED-generated data, every method achieves a clear improvement, with Dice

**Table 5**

Ablation study on diffusion generation strategies for distillation. Incremental performance gains are observed by progressively introducing DANR and SSS, with the full DCED achieving the best segmentation accuracy.

| Data Strategy                  | Diffusion | DANR     | SSS      | Dice (%) $\uparrow$ | HD95 $\downarrow$ |
|--------------------------------|-----------|----------|----------|---------------------|-------------------|
| KD + Base Dataset              | –         | –        | –        | 81.25               | 3.85              |
| KD + Augmented (Pure Diff)     | ✓         | –        | –        | 84.12               | 3.25              |
| KD + Augmented (+ DANR)        | ✓         | ✓        | –        | 88.45               | 2.68              |
| <b>KD + Augmented (+ Full)</b> | <b>✓</b>  | <b>✓</b> | <b>✓</b> | <b>91.63</b>        | <b>2.05</b>       |

gains ranging from 3.94 to 7.05 percentage points. In particular, all distillation-based methods exceed 88% Dice under the *Real + DCED* setting, confirming that the generated samples are broadly beneficial rather than being tailored to a specific distillation objective. Among the existing methods, CWD achieves the highest Dice score of 91.37%, while PRRD obtains the lowest HD95 of 2.18.  $D^3$ -KD performs best under both settings and further improves the results to 91.63% Dice and 2.05 HD95. These results show that combining diffusion-based distribution expansion with the proposed structural distillation strategy leads to more effective knowledge transfer under source-domain-unknown conditions.

**Table 6**

Effect of different noise rescaling strategies. Comparison of segmentation performance under different CFG stabilization schemes.

| Rescaling Strategy                    | Dice (%)↑    | HD95↓       |
|---------------------------------------|--------------|-------------|
| Standard CFG                          | 84.12        | 3.25        |
| Epsilon Scaling ( $\lambda = 1.005$ ) | 86.30        | 2.92        |
| <b>DANR (Ours)</b>                    | <b>88.45</b> | <b>2.68</b> |

### 5.6. Ablation study

To analyze the impact of diffusion generation strategies on cross-domain distillation, we perform a step-wise ablation study of the proposed DCED module. Table 5 reports the final distilled student performance, which directly reflects the effectiveness of diffusion-augmented knowledge transfer.

**Standard Diffusion.** When synthetic samples generated by standard CFG-based diffusion are introduced (Row 2), the student model benefits from an expanded data distribution, achieving a Dice score of 84.12%. However, the improvement remains limited, as prediction drift and sampling instability introduce structural artifacts that partially misalign with the teacher’s decision boundary, thereby constraining effective knowledge transfer.

**Effect of DANR.** By incorporating DANR (Row 3), the variance mismatch and magnitude overestimation induced by CFG are explicitly corrected. This stabilizes the diffusion trajectory and yields synthetic samples that better preserve task-relevant structural cues. As a result, the student model can more reliably exploit the teacher’s supervision, leading to a marked performance gain to 88.45%.

**Effect of SSS.** The full DCED configuration (Row 4) further integrates Shifted-Start Sampling, which avoids low-SNR early timesteps where prediction errors are prone to amplification. By jointly improving distribution coverage and suppressing error accumulation, DCED provides a higher-fidelity distributional bridge between the target domain and the teacher’s latent knowledge. Consequently, the distilled student achieves the best overall performance, with a Dice score of 91.63% and the lowest HD95 of 2.05.

Overall, this ablation study demonstrates that diffusion-based data augmentation is most effective for source-domain-unknown knowledge distillation only when sampling drift is explicitly controlled and early-stage instability is mitigated.

### 5.7. Comparison with epsilon scaling

We further compare the proposed DANR with a representative Epsilon Scaling stabilization strategy (Ning et al., 2024). Following the original setting, Epsilon Scaling rescales the guided noise by dividing it with a fixed factor  $\lambda > 1$ , which is equivalent to uniformly shrinking the predicted noise magnitude as  $\epsilon_{\text{scaled}} = \epsilon_{\text{cfg}}/\lambda$ . For a fair comparison, we apply the same operation directly to the CFG output  $\epsilon_{\text{cfg}}$  using  $\lambda = 1.005$ .

As reported in Table 6, epsilon scaling yields moderate performance improvements over standard CFG, indicating that simple global magnitude suppression can partially alleviate sampling instability. However, its effectiveness is limited due to the lack of drift awareness and timestep adaptivity. Unlike conventional guidance scaling strategies that mainly focus on improving diffusion sampling quality, DANR is designed to suppress prediction drift for stable supervision generation under source-free knowledge distillation settings. Accordingly, DANR further improves both Dice and HD95 by explicitly calibrating the guided noise based on its relative variance, demonstrating superior robustness against multi-step error accumulation.

**Table 7**

Comparison between DDIM with fewer steps and SSS under similar effective sampling steps.

| Sampling Strategy | Effective Steps | Time / img (s)↓ | Dice (%)↑    | HD95↓         |
|-------------------|-----------------|-----------------|--------------|---------------|
| DDIM              | 160             | <b>3.75</b>     | 82.16        | 3.7854        |
| <b>SSS</b>        | 160             | 3.81            | <b>87.36</b> | <b>2.8123</b> |
| DDIM              | 170             | <b>3.97</b>     | 82.87        | 3.6210        |
| <b>SSS</b>        | 170             | 4.04            | <b>88.25</b> | <b>2.4512</b> |
| DDIM              | 180             | <b>4.17</b>     | 83.42        | 3.4821        |
| <b>SSS</b>        | 180             | 4.25            | <b>87.25</b> | <b>2.6845</b> |
| DDIM              | 190             | <b>4.40</b>     | 83.86        | 3.3456        |
| <b>SSS</b>        | 190             | 4.49            | <b>85.87</b> | <b>3.0215</b> |

**Table 8**

Comparison with different diffusion sampling solvers under 170 effective sampling steps.

| Sampling Strategy | Effective Steps | Time / img (s)↓ | Dice (%)↑    | HD95↓       |
|-------------------|-----------------|-----------------|--------------|-------------|
| DDIM              | 170             | <b>3.97</b>     | 82.87        | 3.62        |
| DPM-solver        | 170             | 4.02            | 85.34        | 3.12        |
| DPM-solver + +    | 170             | 4.05            | 86.68        | 2.86        |
| <b>Ours (SSS)</b> | 170             | 4.04            | <b>88.25</b> | <b>2.45</b> |

### 5.8. Comparison of shifted-start sampling and DDIM under matched sampling steps

Table 7 presents a controlled comparison between the proposed SSS and DDIM under similar effective sampling steps. By aligning the number of denoising steps, this experiment provides a fair evaluation of generation quality and efficiency.

As shown in the table, SSS consistently achieves significantly higher Dice scores and lower HD95 values than DDIM across all configurations, demonstrating its superiority. Notably, the best performance is obtained at an effective step of 170 (corresponding to a shift of 30), where SSS reaches the highest Dice score of 88.25% and the lowest HD95 of 2.4512.

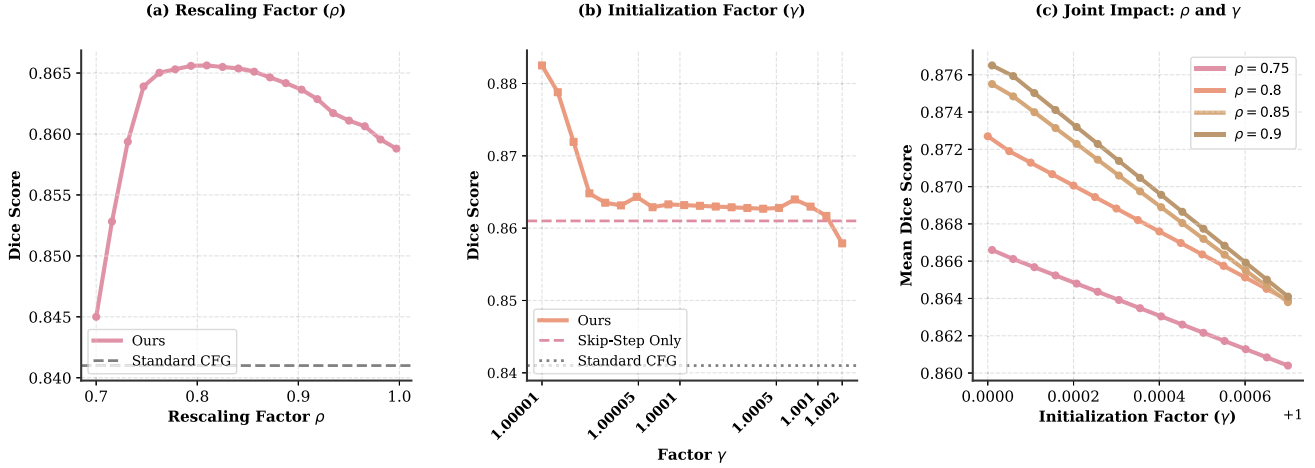
In terms of efficiency, although SSS introduces a slight increase in per-image inference time compared to DDIM with the same effective steps, the performance gain is substantial and consistent. This indicates that the advantage of SSS does not arise from reducing the number of sampling steps, but from avoiding early low-SNR stages that are prone to prediction drift.

When the shift becomes too large (e.g., effective steps of 160), a slight degradation in performance can be observed, suggesting that excessive skipping may discard useful denoising stages. Overall, these results validate that a moderate shifted initialization provides a better trade-off between efficiency and reconstruction quality.

### 5.9. Comparison with different sampling solvers

We further compare the proposed SSS strategy with advanced diffusion sampling solvers, including DPM-solver and DPM-solver + +, under the same effective sampling steps. As shown in Table 8, both DPM-solver variants improve over vanilla DDIM, indicating that more advanced ODE-based solvers can partially alleviate sampling errors.

However, the proposed SSS strategy still achieves the best overall performance in terms of Dice and HD95, while maintaining comparable inference efficiency. This suggests that the advantage of SSS does not solely arise from faster numerical solvers, but from explicitly avoiding early low-SNR stages that are prone to prediction drift in the SFDA-KD setting. These results suggest that the superiority of SSS does not merely arise from reducing the number of denoising steps. Instead, its advantage mainly stems from avoiding instability-dominated early diffusion stages, which improves the reliability of generated supervision during knowledge distillation.



**Fig. 6.** Sensitivity and interaction analysis of sampling hyperparameters. (a) **impact of rescaling factor ( $\rho$ )**: the inverted U-shaped curve shows that noise rescaling independently improves performance over the baseline. (b) **Impact of initialization factor ( $\gamma$ )**: a precise micro-adjustment to  $\gamma = 1.00001$  yields a substantial performance leap (dice > 0.87). (c) **Joint interaction analysis**: while the pure initialization strategy reaches a higher peak mean dice, the combination strategies (e.g.,  $\rho = 0.9$ ) exhibit smoother performance decay across the parameter space, suggesting enhanced robustness against initialization variance.

**Table 9**

Ablation study on the effect of sampling steps. The best overall performance is achieved with 200 sampling steps, which is adopted in subsequent experiments. HD95 values in this table are reported in the original pixel coordinate scale.

| Sampling Steps | Dice (%) $\uparrow$ | IoU (%) $\uparrow$ | HD95 $\downarrow$ |
|----------------|---------------------|--------------------|-------------------|
| 100            | 82.13               | 74.01              | 29.33             |
| <b>200</b>     | <b>82.27</b>        | <b>74.14</b>       | <b>28.57</b>      |
| 500            | 81.90               | <b>74.35</b>       | 31.18             |
| 1000           | 81.74               | 73.55              | 29.40             |

#### 5.10. Parameter sensitivity analysis

Beyond the cumulative improvements reported in Table 5, we further analyze the intrinsic sensitivity of key sampling hyperparameters in Fig. 6 to better understand their individual roles and interactions.

As shown in Fig. 6(a), the rescaling factor  $\rho$  exhibits a clear inverted U-shaped trend. Moderate rescaling substantially improves generation quality over standard CFG, while overly aggressive scaling leads to gradual performance degradation, indicating that excessive noise amplification reintroduces sampling instability. This observation supports our design motivation that magnitude calibration is beneficial only within a controlled range.

Fig. 6(b) further reveals that the initialization factor  $\gamma$  is highly sensitive near unity. A precise micro-adjustment around  $\gamma = 1.00001$  yields a pronounced performance peak (Dice > 0.87), suggesting that slight variance inflation at the shifted starting point can effectively compensate for distribution mismatch induced by skipping early timesteps. However, this sharp optimum also indicates limited robustness when  $\gamma$  is used alone.

The joint analysis in Fig. 6(c) highlights the complementary roles of  $\rho$  and  $\gamma$ . While pure initialization achieves a higher peak Dice score, the combined strategy consistently demonstrates smoother performance decay across the parameter space. This behavior suggests enhanced robustness to initialization variance and sampling noise, which is particularly desirable for stable diffusion-based augmentation in downstream distillation.

Based on these observations, we adopt the combined configuration in the final DCED framework, prioritizing robustness and stability over narrowly optimized peak performance.

#### 5.11. Effect of sampling steps

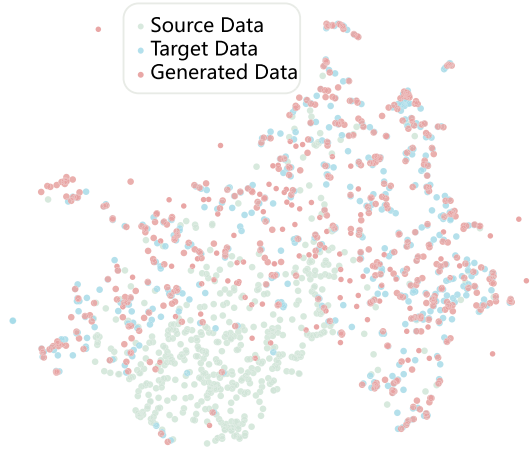
We study the impact of the number of diffusion sampling steps on the final segmentation performance, as shown in Table 9. Interestingly, reducing the number of sampling steps does not degrade performance. Instead, using 200 steps achieves the best overall results in terms of Dice and HD95. This observation suggests that excessively long sampling trajectories may introduce accumulated noise and prediction drift, which can negatively affect the quality of generated samples. In contrast, a moderate number of steps strikes a better balance between generation stability and sample fidelity. For a fair comparison, the proposed SSS strategy adopts a consistent shift ratio of 15% across all sampling configurations. We also observe that while larger step counts (e.g., 1000 steps) slightly improve performance on high-confidence subsets, they do not translate into better overall results. This further indicates that overly long diffusion processes may overfit to easier samples while failing to generalize across the full data distribution. Considering both performance and computational efficiency, we adopt 200 sampling steps as the default setting in all experiments.

Together with comparisons involving guidance mechanisms (e.g., CFG and epsilon scaling), sampling schedules (SSS), and DDIM-based sampling, our experiments provide a comprehensive analysis of the diffusion sampling process.

#### 5.12. Distribution bridging analysis via UMAP visualization

For analysis purposes only, we additionally access source-domain samples to visualize feature distributions and compute distributional statistics. These source-domain samples are used exclusively for post-hoc analysis and are not involved in diffusion generation or knowledge distillation training.

To further investigate whether the generated samples effectively reduce the distribution gap between domains, we visualize the feature distributions of source data, target data, and diffusion-generated samples using UMAP. As illustrated in Fig. 7, the source and target data exhibit a noticeable distribution gap, forming partially separated clusters in the feature space. In contrast, the generated data, constructed by combining real target samples and generated samples, spans the intermediate regions between the two domains. This indicates that the generated samples do not merely increase data diversity, but instead provide a meaningful interpolation that connects the source and target distributions. Such behavior can be understood from the formulation of



**Fig. 7.** UMAP visualization of feature distributions for source data (BUS-BRA), target data (BUS-UCLM), and generated data. The generated data spans the regions of both domains, showing increased coverage of the feature space.

**Table 10**

Comparison of distribution discrepancies with respect to the source distribution using MMD and FID.

| Data Distribution    | MMD↓          | FID↓         |
|----------------------|---------------|--------------|
| Target vs. Source    | 1.2632        | 51.91        |
| Generated vs. Source | <b>1.0745</b> | <b>22.56</b> |

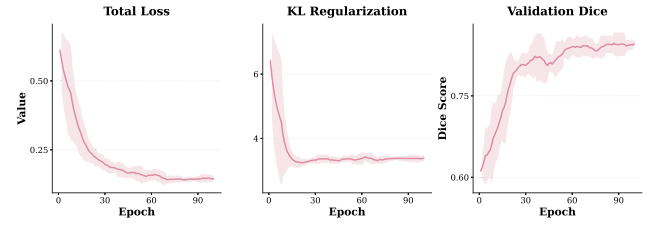
the augmented distribution  $P'_T = (1 - \lambda)P_T + \lambda P_G$ , where the generated data  $P_G$  serves as a bridge to expand the coverage of the target domain. By filling the low-density regions between domains, the generated data effectively reduces the domain discrepancy and facilitates more stable knowledge transfer from the teacher model.

#### 5.13. Distribution alignment analysis via MMD and FID

To quantitatively evaluate whether the generated data reduce the distribution discrepancy with respect to the source domain, we employ Maximum Mean Discrepancy (MMD) and Fréchet Inception Distance (FID). The two metrics characterize distribution discrepancy from complementary perspectives, with lower values indicating greater similarity to the source distribution. As shown in Table 10, the MMD between the generated and source distributions is reduced from 1.2632 to 1.0745 compared with the original target distribution. A more pronounced reduction is also observed for FID, which decreases from 51.91 for the target–source pair to 22.56 for the generated–source pair.

These consistent reductions indicate that the generated samples are closer to the source distribution than the original target-domain samples in the evaluated feature space. This quantitative observation is consistent with the UMAP visualization in Fig. 7, where the generated samples extend beyond the original target-domain cluster and occupy intermediate regions between the target and source domains. Together, the MMD, FID, and UMAP results provide complementary quantitative and qualitative evidence that the proposed diffusion-based generation expands the coverage of the target distribution and partially bridges the domain discrepancy.

Overall, these results suggest that the performance improvement is not solely attributable to an increase in training sample quantity or diversity. Instead, the generated data exhibit improved distributional proximity to the source domain, providing empirical support for our coverage-oriented generation assumption and the proposed distribution-bridging mechanism.



**Fig. 8.** Training dynamics of the student network. **(Left) total loss:** rapid convergence indicates stable optimization of the composite objective. **(Middle) KL regularization:** the steady decline confirms effective alignment between student and teacher distributions. **(Right) validation dice:** continuous improvement demonstrates that knowledge transfer directly translates into segmentation performance.

#### 5.14. Training curves analysis

We validate the optimization stability of our framework by monitoring key training metrics over 100 epochs. Fig. 8 illustrates the evolution of Total Loss, KL Regularization, and Validation Dice. The training curves demonstrate stable and effective optimization. The total loss exhibits smooth and rapid convergence, indicating well-balanced supervision and numerical stability. Meanwhile, the steady reduction of KL regularization accompanied by consistent improvement in validation Dice confirms that the student model effectively aligns with the teacher while achieving strong generalization on the target domain.

### 6. Conclusion and future work

In this work, we investigate knowledge distillation under the *source-domain-unknown* setting, a practical yet underexplored scenario in medical image segmentation where the teacher’s training data are inaccessible. Focusing on breast ultrasound segmentation, we provide a theoretical analysis showing that cross-domain knowledge distillation remains feasible under domain mismatch, but its effectiveness is fundamentally constrained by the *training-sampling gap* induced by limited target-domain coverage. To address this limitation, we propose D<sup>3</sup>-KD, a diffusion-enhanced distillation paradigm that expands the support of the target-domain distribution and increases its potential overlap with the unknown teacher domain. Moreover, we identify prediction drift as a critical bottleneck in diffusion-based augmentation and introduce a DCED framework to enable stable and efficient data generation. Extensive experiments on breast ultrasound segmentation demonstrate consistent performance improvements under domain discrepancies, validating both the theoretical insights and the practical value of source-domain-unknown knowledge distillation.

Future work will explore extending the proposed framework to more complex segmentation tasks and additional medical imaging modalities. Incorporating task-aware or uncertainty-guided diffusion strategies may further improve the relevance of generated samples to the teacher’s decision boundary. In addition, developing more adaptive sampling schemes and integrating source-domain-unknown distillation with continual or foundation-model-based learning paradigms may further enhance robustness and scalability in real-world clinical applications.

#### CRedit authorship contribution statement

**Jiati Cai:** Writing – original draft, Methodology, Conceptualization; **Kun Wang:** Visualization, Validation, Methodology; **Meihui Zhong:** Validation, Methodology; **Haohan Zhang:** Visualization, Investigation; **Yuxin Tong:** Validation, Data curation; **Junren Wang:** Methodology, Data curation; **Fan Zhou:** Supervision, Conceptualization; **Jing Jing:** Visualization, Supervision; **Jin Yin:** Supervision, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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